MATH 4334
Numerical Analysis
Lecture Notes

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Chapter 1

Approximation and Taylor's Theorem

1.1 Numerical problems and error

Numerical analysis studies methods for solving problems involving continuous variables by finite computations. The computed answer is usually an approximation. Along with an approximation, we want a bound on its error.

Example 1.1. Consider

$$x^5 - 4x^4 + x^3 - x^2 + 2x - 1 = 0.$$

An odd-degree real polynomial has a real zero, but existence alone does not give a practical way to compute it. A numerical method produces approximations to a zero and estimates their accuracy.

Exercise 1.1. Find an approximation to a real zero of $P(x) = x^5 - 4x^4 + x^3 - x^2 + 2x - 1$ with absolute error at most 10^{-4}.

Solution. Evaluate the polynomial in nested form,

$$P(x) = ((((x - 4)x + 1)x - 1)x + 2)x - 1.$$

For example, a search starting with $P(3) = -58$ and $P(4) = 55$ leads to the narrower interval whose endpoint values are

$$P(3.7728) = -0.01869307466069573632, \qquad P(3.7729) = 0.00040491292351839649.$$

These terminating decimal evaluations are exact. Since P is continuous, it has a zero $r \in (3.7728, 3.7729)$. The midpoint $\widetilde{r} = 3.77285$ therefore satisfies

$$|r - \widetilde{r}| < \frac{3.7729 - 3.7728}{2} = 5 \times 10^{-5} < 10^{-4}.$$

Example 1.2. Let $I = \int_0^{\pi/2} e^x \sin x \, dx$. Integration by parts twice gives

$$I = [-e^x \cos x]_0^{\pi/2} + \int_0^{\pi/2} e^x \cos x \, dx,$$

$$\int_0^{\pi/2} e^x \cos x \, dx = [e^x \sin x]_0^{\pi/2} - I.$$

Thus $2I = [e^x(\sin x - \cos x)]_0^{\pi/2} = e^{\pi/2} + 1$, and

$$I = \frac{e^{\pi/2} + 1}{2}.$$

The calculation also illustrates why a numerical procedure needs a specified rule: even a familiar symbolic operation requires a choice of how to proceed.

Exercise 1.2. Evaluate $\int x \ln x \, dx$ for $x > 0$.

Solution. Choose $u = \ln x$ and $dv = x\,dx$. Then $du = dx/x$ and $v = x^2/2$, so

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C.$$

Differentiating the right side gives $x \ln x$.

Definition 1.3. If x is an exact value and $\widetilde{x}$ is an approximation, the *absolute error* is $|x - \widetilde{x}|$. For $x \neq 0$, the *relative error* is $|x - \widetilde{x}|/|x|$.

1.2 Taylor polynomials and remainders

Theorem 1.4 (Taylor's theorem)**.** *Suppose $f \in C^n([a, b])$ and $f^{(n+1)}$ exists on (a, b). For $c, x \in [a, b]$, $x \neq c$, there is a point ξ strictly between c and x such that*

$$f(x) = T_n(x) + R_n(x), \qquad T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x - c)^k, \qquad R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - c)^{n+1}.$$

Exercise 1.3. Prove Taylor's theorem using Rolle's theorem.

Solution. Fix $x \neq c$ and put

$$K = \frac{f(x) - T_n(x)}{(x - c)^{n+1}}, \qquad F(t) = f(t) - T_n(t) - K(t - c)^{n+1}.$$

Then $F(x) = F(c) = 0$, and $F^{(j)}(c) = 0$ for $1 \leq j \leq n$. Repeated application of Rolle's theorem gives a point ξ between c and x with $F^{(n+1)}(\xi) = 0$. For $n = 0$, one application suffices. For $n \geq 1$, the first application gives a zero of F' between c and x; together with $F'(c) = 0$, this gives a zero of F'', and the process continues until order $n + 1$. Consequently

$$0 = f^{(n+1)}(\xi) - (n + 1)!K.$$

Substitution of K yields the stated remainder.

If $|f^{(n+1)}(t)| \leq M$ between c and x, then

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x - c|^{n+1}.$$

With $c = 0$, T_n is a Maclaurin polynomial. For familiar analytic functions,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \qquad (x \in \mathbb{R}),$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \qquad (x \in \mathbb{R}),$$
$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \qquad (-1 < x < 1).$$

Equivalently, $\ln(1-x) = -x - x^2/2 - x^3/3 - \cdots$ for $|x| < 1$.

Note 1.5. The existence of derivatives of every order does not by itself imply that a function equals its Taylor series. Taylor's theorem gives a finite expansion; equality with the infinite series requires $R_n(x) \to 0$ as $n \to \infty$.

Example 1.6. For $f(x) = \cos x$ and $c = 0$, $T_2(x) = 1 - x^2/2$. Hence

$$\cos(0.01) \approx 1 - \frac{(0.01)^2}{2} = 0.99995.$$

The degree-two remainder satisfies

$$|R_2(0.01)| = \frac{|\sin \xi|}{6}(0.01)^3 \le \frac{10^{-6}}{6} \approx 1.667 \times 10^{-7}.$$

Since $f'''(0) = 0$, $T_3 = T_2$. Applying the degree-three remainder instead gives the sharper bound

$$|\cos(0.01) - 0.99995| \le \frac{(0.01)^4}{24} \approx 4.167 \times 10^{-10}.$$

The polynomial is the same; the second estimate uses one more derivative.

1.3 Order notation

Taylor's theorem about a point x can also be written

$$f(x+h) = \sum_{k=0}^{n} \frac{f^{(k)}(x)}{k!} h^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} h^{n+1}.$$

The intermediate point depends on h, but a uniform derivative bound controls the remainder.

For $c < x$ and $h > 0$, respectively, the intermediate points lie as follows:



Definition 1.7. We write $E(h) = O(h^p)$ as $h \to 0$ if there are constants $C, \eta > 0$ such that $|E(h)| \le C|h|^p$ whenever $0 < |h| < \eta$. More generally, $f = O(g)$ means $|f| \le C|g|$ for some constant $C > 0$ sufficiently near the specified limit. If g is nonzero there, this is equivalent to boundedness of $|f/g|$.

Thus a bounded $(n+1)$st derivative gives a Taylor remainder $O(h^{n+1})$. This is an upper bound on the error, not a claim that the error is always a nonzero constant times h^{n+1}.

Chapter 2

Number Representation and Floating-Point Arithmetic

2.1 Positional representations

For an integer base $\beta \geq 2$, the digits are $0, 1, \ldots, \beta - 1$. A positional representation means

$$(a_n a_{n-1} \cdots a_0 . b_1 b_2 \cdots)_\beta = \sum_{k=0}^{n} a_k \beta^k + \sum_{k=1}^{\infty} b_k \beta^{-k}.$$

For example,

$$2375 = 2 \cdot 10^3 + 3 \cdot 10^2 + 7 \cdot 10 + 5, \qquad 0.2543 = 2 \cdot 10^{-1} + 5 \cdot 10^{-2} + 4 \cdot 10^{-3} + 3 \cdot 10^{-4}.$$

Decimal expansions such as those of π, e, and $1/3$ need not terminate.

Base	Name	Digits
10	Decimal	$0, \ldots, 9$
2	Binary	$0, 1$
8	Octal	$0, \ldots, 7$
16	Hexadecimal	$0, \ldots, 9, A, B, C, D, E, F$

Example 2.1. Direct evaluation gives

$$(1011.101)_2 = 2^3 + 2 + 1 + 2^{-1} + 2^{-3} = 11.625.$$

For a repeating expansion, let $x = (0.\overline{1011})_2$. Multiplication by 2^4 shifts one block:

$$16x = (1011.\overline{1011})_2 = 11 + x, \qquad x = \frac{11}{15}.$$

2.1.1 Converting the integer and fractional parts

For a nonnegative integer N, write

$$N = a_0 + \beta\big(a_1 + \beta(a_2 + \cdots + \beta a_n)\big).$$

Successive divisions by β give the remainders $a_0, a_1, \ldots, a_n$. Read them in reverse order to obtain the digits. The nested expression is also useful for evaluating polynomials.

For $0 \le x < 1$, set $r_0 = x$ and compute

$$b_k = \lfloor \beta r_{k-1} \rfloor, \qquad r_k = \beta r_{k-1} - b_k.$$

Then $b_1, b_2, \ldots$ are the fractional digits. At each step $0 \le r_k < 1$, and

$$x = \sum_{j=1}^{k} b_j \beta^{-j} + r_k \beta^{-k}.$$

The truncation error satisfies $0 \le r_k \beta^{-k} < \beta^{-k}$ and tends to zero as $k \to \infty$; the remainders r_k themselves need not tend to zero.

Example 2.2. Successive binary divisions of 135 give

$$\begin{aligned} 135 &= 2(67) + 1, & 67 &= 2(33) + 1, \\ 33 &= 2(16) + 1, & 16 &= 2(8) + 0, \\ 8 &= 2(4) + 0, & 4 &= 2(2) + 0, \\ 2 &= 2(1) + 0, & 1 &= 2(0) + 1. \end{aligned}$$

Hence $(135)_{10} = (10000111)_2$. For the fraction 0.273, the first multiplications are

$$\begin{aligned} 2(0.273) &= 0.546 \Rightarrow b_1 = 0, \\ 2(0.546) &= 1.092 \Rightarrow b_2 = 1, \\ 2(0.092) &= 0.184 \Rightarrow b_3 = 0, \\ 2(0.184) &= 0.368 \Rightarrow b_4 = 0, \\ 2(0.368) &= 0.736 \Rightarrow b_5 = 0, \\ 2(0.736) &= 1.472 \Rightarrow b_6 = 1, \\ 2(0.472) &= 0.944 \Rightarrow b_7 = 0. \end{aligned}$$

Thus $(0.273)_{10} = (0.0100010\cdots)_2$.

Exercise 2.1. Explain why the binary expansion of 0.273 is nonterminating but eventually periodic.

Solution. In lowest terms $0.273 = 273/1000$ and $1000 = 2^3 \cdot 5^3$. A terminating binary fraction has denominator dividing a power of 2, so this fraction cannot terminate. For a rational number with denominator 1000, the fractional remainders in repeated multiplication have the form $j/1000$, $0 \le j < 1000$. There are finitely many possibilities, so a remainder repeats. All subsequent digits then repeat as well.

2.1.2 Octal and hexadecimal as intermediate bases

Each octal digit is a group of three binary digits; each hexadecimal digit is a group of four. Group outward from the radix point, adding zeros at the outer ends if necessary.

Octal	0	1	2	3	4	5	6	7
Binary	000	001	010	011	100	101	110	111

Example 2.3. Since $135 = 8(16) + 7$, $16 = 8(2) + 0$, and $2 = 8(0) + 2$,

$$(135)_{10} = (207)_8 = (010\,000\,111)_2 = (10000111)_2.$$

For 75.75, the integer divisions are $75 = 8(9) + 3$, $9 = 8(1) + 1$, $1 = 8(0) + 1$; the fractional multiplication is $8(0.75) = 6$. Therefore

$$(75.75)_{10} = (113.6)_8 = (001\,001\,011.110)_2 = (1001011.11)_2.$$

For hexadecimal,

$$(2A)_{16} = 2(16) + 10 = 42, \qquad (135)_{10} = (87)_{16} = (1000\,0111)_2,$$
$$(75)_{10} = (4B)_{16} = (0100\,1011)_2.$$

2.2 Normalized numbers and a finite machine

A nonzero real number can be written in decimal normalized form

$$x = \pm(0.d_1d_2\cdots)_{10}10^m, \qquad d_1 \neq 0, \quad m \in \mathbb{Z}.$$

For instance,

$$543.26511 = 0.54326511 \times 10^3,$$
$$0.000343728 = 0.343728 \times 10^{-3},$$
$$17832411.23004 = 0.1783241123004 \times 10^8.$$

In binary one may use either $\pm(0.1b_2\cdots)_2 2^m$ or, equivalently, $\pm(1.b_2b_3\cdots)_2 2^{m-1}$. A machine retains only finitely many significand digits and a bounded exponent range. Exactly representable real values are called machine numbers. For example, $1/2$ terminates in binary, whereas $1/3$ does not.

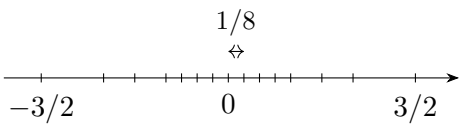
Example 2.4 (A small machine)**.** Consider all encodings

$$\pm(0.b_1b_2)_2\, 2^m, \qquad b_1, b_2 \in \{0, 1\}, \quad m \in \{-1, 0, 1\},$$

without requiring normalization. There are $2 \cdot 2 \cdot 2 \cdot 3 = 24$ encodings, but some represent the same value. The nonnegative values are

$$0, \quad \frac{1}{8}, \quad \frac{1}{4}, \quad \frac{3}{8}, \quad \frac{1}{2}, \quad \frac{3}{4}, \quad 1, \quad \frac{3}{2}.$$

For example, $(0.01)_2 2^0 = (0.10)_2 2^{-1} = 1/4$. Including negatives gives 15 distinct real numbers. Magnitudes above $3/2$ cannot be represented; values with magnitude below $1/8$ also have no nonzero exact representation in this toy system. Rounding to a nearby representable value introduces error.



The spacing is not uniform across the full range: larger magnitudes have larger gaps.

2.3 Binary floating-point fields

The binary32 (single) format has one sign bit, eight exponent bits, and 23 stored fraction bits. Binary64 (double) has one sign bit, eleven exponent bits, and 52 stored fraction bits. An 80-bit x86 extended format uses one sign bit, fifteen exponent bits, and 64 significand bits, including an explicit integer bit; the meaning of a programming language's `long double` depends on the implementation.[1]

For binary32, let $s \in \{0,1\}$ be the sign, c the unsigned exponent field, and $F = (0.b_1 \cdots b_{23})_2$. A normal value is

$$(-1)^s (1+F) 2^{c-127}, \qquad 1 \le c \le 254.$$

The leading 1 is implicit. Thus there are 24 significant binary digits and the unbiased exponent ranges from -126 to 127. Special exponent fields are interpreted as follows:

Exponent field	Fraction	Value
$c = 0$	$F = 0$	Signed zero
$c = 0$	$F \neq 0$	$(-1)^s F 2^{-126}$ (subnormal)
$c = 255$	$F = 0$	Signed infinity
$c = 255$	$F \neq 0$	NaN (not a number)

The largest significand follows from a finite geometric sum:

$$1 + \sum_{j=1}^{23} 2^{-j} = 1 + \frac{\frac{1}{2}(1 - 2^{-23})}{1 - \frac{1}{2}} = 2 - 2^{-23}.$$

Thus the largest finite positive binary32 number is

$$(2 - 2^{-23}) 2^{127} \approx 3.4028235 \times 10^{38}.$$

The smallest positive normal number is $2^{-126} \approx 1.1754944 \times 10^{-38}$; the smallest positive subnormal number is 2^{-149}. Subnormal values extend the range toward zero with fewer significant digits.

Signed zero has sign field 0 or 1 and all other bits zero. In arithmetic with the usual nontrapping interpretation, $1/(+0) = +\infty$ and $1/(-0) = -\infty$. Invalid operations such as $0/0$, ∞/∞, $0 \cdot \infty$, and $\infty - \infty$ produce NaN.

Example 2.5 (Decoding a bit pattern)**.** The fields

$$\underbrace{0}_{s} \quad \underbrace{10001011}_{c} \quad \underbrace{10100000000000000000000}_{\text{fraction}}$$

give $s = 0$, $c = 139$, $c - 127 = 12$, and $(1.101)_2 = 1 + 1/2 + 1/8 = 1.625$. The represented value is $1.625\, 2^{12} = 6656$. Replacing the first bit by 1 changes the value to -6656.

Example 2.6 (Encoding a number)**.** For $x = -52.234375$, $52 = (64)_8 = (110100)_2$ and $0.234375 = 15/64 = (0.17)_8 = (0.001111)_2$. Thus

$$x = -(110100.001111)_2 = -(1.10100001111)_2\, 2^5.$$

The sign is 1, the exponent field is $5 + 127 = 132 = (204)_8 = (10000100)_2$, and the stored fraction is 10100001111000000000000. The fields are

$$1 \quad 10000100 \quad 10100001111000000000000.$$

This value is exactly representable.

[1] For the storage formats and special values, see Oracle, *Numerical Computation Guide*, Section 2.2, IEEE Formats.

2.4 Rounding error and machine epsilon

In binary32, the next machine number above 1 is $1 + 2^{-23}$. We call

$$\varepsilon = 2^{-23} \approx 1.1920929 \times 10^{-7}$$

the machine epsilon. The unit roundoff for rounding to nearest is $u = \varepsilon/2 = 2^{-24}$.

Suppose a positive real number has the form $x = q2^m$, $1 \le q < 2$, $m \in \mathbb{Z}$, between the smallest positive normal and the largest finite binary32 number. If x is exactly representable, rounding leaves it unchanged. Otherwise let $x_- < x < x_+$ be the consecutive binary32 numbers bracketing it. Their spacing is $x_+ - x_- = 2^{m-23}$. Rounding to nearest, denoted by fl, gives

$$|x - \mathrm{fl}(x)| \le \tfrac{1}{2}(x_+ - x_-) = 2^{m-24},$$

and therefore

$$\frac{|x - \mathrm{fl}(x)|}{|x|} \le \frac{2^{m-24}}{q2^m} \le 2^{-24} = u.$$

A tie can be resolved by choosing the even final significand bit.

Example 2.7. Using octal as an intermediate base gives $9 = (11)_8 = (1001)_2$ and $0.4 = (0.\overline{3146})_8 = (0.\overline{0110})_2$. Thus $x = 9.4 = (1.00101100110011\cdots)_2\, 2^3$, and the binary32 spacing is 2^{-20}. Its neighboring values satisfy

$$x - x_- = 0.4\, 2^{-20}, \qquad x_+ - x = 0.6\, 2^{-20}.$$

Consequently

$$\mathrm{fl}(9.4) = x_- = 9.3999996185302734375,$$

$$|x - \mathrm{fl}(x)| = 0.4\, 2^{-20}, \qquad \frac{|x - \mathrm{fl}(x)|}{9.4} = \frac{16}{47}\varepsilon < u.$$

Even when $9.4 - 9 - 0.4 = 0$ in exact arithmetic, storing the decimal inputs in binary32 changes them. With every operation rounded to binary32,

$$\mathrm{fl}\big(\mathrm{fl}\big(\mathrm{fl}(9.4) - 9\big) - \mathrm{fl}(0.4)\big) = -3.25\varepsilon = -3.874301910400390625 \times 10^{-7}.$$

Here $\mathrm{fl}(0.4) = 0.40000000596046447753906 25$. Rounding of the input 0.4 matters as well.

2.4.1 A model for arithmetic

For rounding to nearest, suppose the exact real quantity z being rounded is zero or satisfies

$$2^{-126} \le |z| \le (2 - 2^{-23})2^{127}.$$

Then we use

$$\mathrm{fl}(z) = z(1 + \delta), \qquad |\delta| \le u.$$

At $z = 0$ we may take $\delta = 0$. This range assumption applies separately to every rounding below, including input conversions and intermediate results. An exactly represented subnormal input has no conversion error and may still be used if the results being rounded satisfy the assumption. For a real input in the subnormal range, rounding to nearest with gradual underflow instead guarantees the

absolute bound $|\,\mathrm{fl}(z) - z| \le 2^{-150}$; the relative bound $|\delta| \le u$ need not hold. For example, $z = 2^{-151}$ rounds to zero and has relative error 1. In particular, for machine inputs and $\circ \in \{+, -, \times, /\}$,

$$\mathrm{fl}(x \circ y) = (x \circ y)(1 + \delta).$$

Division requires $y \neq 0$. If the real inputs must first be rounded, then

$$\mathrm{fl}\big(\mathrm{fl}(x) \circ \mathrm{fl}(y)\big) = \big[x(1 + \delta_1) \circ y(1 + \delta_2)\big](1 + \delta_3).$$

There are separate errors for the two inputs and for the operation.

For machine values x, y, z, computing $x(y + z)$ gives

$$\begin{aligned} s = \mathrm{fl}(y + z) &= (y + z)(1 + \delta_1), \\ \mathrm{fl}(xs) &= x(y + z)(1 + \delta_1)(1 + \delta_2). \end{aligned}$$

Thus the combined relative perturbation has magnitude at most $2u + u^2$. To first order it is at most $2u = \varepsilon$; the product $\delta_1\delta_2$ should be retained for an exact bound.

2.5 Cancellation and loss of significance

For close positive quantities $x \neq y$, a subtraction may magnify errors already present in the inputs. Indeed,

$$\frac{|(x - y) - (\widetilde{x} - \widetilde{y})|}{|x - y|} \le \frac{|x - \widetilde{x}| + |y - \widetilde{y}|}{|x - y|}.$$

The denominator can be very small. This affects, for instance, the difference quotient $[f(a + h) - f(a)]/h$ when h is small and nonzero.

Example 2.8. On a three-significant-digit decimal machine,

$$\sqrt{9.01} \approx 3.00, \qquad \sqrt{9.01} - 3 \approx 0.$$

The exact difference is about 1.6662×10^{-3}. Rationalization gives

$$\sqrt{9.01} - 3 = \frac{0.01}{\sqrt{9.01} + 3} \approx \frac{0.01}{6.00} = 1.67 \times 10^{-3},$$

which avoids subtracting nearly equal numbers.

Example 2.9. For $x^2 + 9^{12}x - 3 = 0$, the quadratic formula gives

$$x_{\pm} = \frac{-9^{12} \pm \sqrt{9^{24} + 12}}{2}.$$

The negative root is about -2.8243×10^{11}. Direct evaluation of the positive root subtracts two numbers of nearly the same size. Instead,

$$x_{+} = \frac{6}{9^{12} + \sqrt{9^{24} + 12}} \approx 1.06221185 \times 10^{-11}.$$

More generally, for real coefficients with $a \neq 0$ and $D = b^2 - 4ac \ge 0$, rationalization gives

$$\frac{-b + \sqrt{D}}{2a} = \frac{(-b + \sqrt{D})(-b - \sqrt{D})}{2a(-b - \sqrt{D})} = \frac{-2c}{b + \sqrt{D}}$$

when the denominator is nonzero. Choose the form that avoids cancellation for the given signs.

2.5.1 Counting displaced binary digits

Suppose $0 < y < x$ and set $r = 1 - y/x$. Write $x = s2^n$ with $n \in \mathbb{Z}$ and $1/2 \le s < 1$. Then $x - y = sr2^n$. If

$$2^{-p} \le r \le 2^{-q}, \qquad p, q \in \mathbb{Z}, \quad 0 \le q \le p,$$

then $2^{-(p+1)} \le sr < 2^{-q}$. Renormalizing the difference to a significand in $[1/2, 1)$ therefore shifts its exponent downward by between q and p positions.

Note 2.10. This shift describes the leading positions canceled by subtraction. It does not say that subtracting two exact machine inputs necessarily introduces a new rounding error. The difficulty is that errors in approximated inputs can be exposed in the remaining digits.

Example 2.11. For $x = 37.593621$ and $y = 37.584216$,

$$1 - \frac{y}{x} \approx 0.0002501754, \qquad 2^{-12} \approx 0.0002441406 < 1 - \frac{y}{x} < 2^{-11} \approx 0.0004882813.$$

Thus the cancellation displaces between 11 and 12 leading binary positions.

Exercise 2.2. For $f(x) = e^x - e^{-2x}$, derive a formula suitable near zero. Find when the ratio of the absolute difference to the larger exponential is at least $1/2$, giving a sufficient condition that at most one leading binary position is displaced.

Solution. Subtracting the Taylor expansions gives

$$f(x) = \sum_{n=1}^{\infty} \frac{1 - (-2)^n}{n!} x^n = 3x - \frac{3}{2}x^2 + \frac{3}{2}x^3 - \frac{5}{8}x^4 + \cdots .$$

Alternatively,

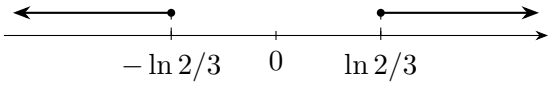
$$f(x) = e^{-2x}(e^{3x} - 1) = e^{-2x}\left(3x + \frac{(3x)^2}{2!} + \frac{(3x)^3}{3!} + \cdots\right).$$

Either series avoids subtracting two exponentials close to 1. For instance, at $x = 0.001$ the leading terms give approximately 0.0029985015. An implementation may evaluate $e^{3x} - 1$ with an accurate small-argument routine.

For $x > 0$, the larger exponential is e^x, so

$$\frac{e^x - e^{-2x}}{e^x} = 1 - e^{-3x} \ge \frac{1}{2} \quad \iff \quad x \ge \frac{\ln 2}{3}.$$

For $x < 0$, divide the positive difference $e^{-2x} - e^x$ by e^{-2x} instead. The condition becomes $1 - e^{3x} \ge 1/2$, or $x \le -\ln 2/3$. At $x = 0$ the ratio is zero. Hence the ratio is at least $1/2$ exactly when $|x| \ge \ln 2/3 \approx 0.231049$. Since the ratio is also less than 1, the preceding displacement bound applies with $p = 1$ and $q = 0$.



Chapter 3

Nonlinear Equations

3.1 Iterative approximation and order of convergence

We seek a root of a nonlinear equation

$$f(x) = 0.$$

For example, $x^3 - x + 1 = 0$ is a nonlinear equation. Bisection, fixed-point iteration, and Newton's method construct successive approximations to a root. If $x_n \to p$ and $f(p) = 0$, we define the absolute error by

$$e_n = |x_n - p|.$$

Only finitely many iterates can be computed. We therefore need an estimate of e_n and a stopping criterion.

Definition 3.1. Suppose $x_n \to p$ and $e_n > 0$ for all sufficiently large n. The sequence has order $q \geq 1$ and asymptotic error constant λ if

$$\lim_{n\to\infty} \frac{e_{n+1}}{e_n^q} = \lambda,$$

where $0 < \lambda < 1$ for $q = 1$ and $0 < \lambda < \infty$ for $q > 1$. Order one is called linear convergence, and order two is called quadratic convergence. An estimate $e_{n+1} \leq Ce_n^q$ for all sufficiently large n expresses an upper error bound of order q; for $q > 1$, C need not be less than one.

Suppose $e_{n+1} \leq Ce_n$ with $0 < C < 1$. Repeated substitution gives

$$e_{n+1} \leq Ce_n \leq C^2 e_{n-1} \leq \cdots \leq C^{n+1} e_0.$$

Thus a stepwise contraction gives a geometric error bound. The converse need not hold: a bound $e_n \leq AC^n$ does not by itself imply $e_{n+1} \leq Ce_n$.

3.2 The bisection method

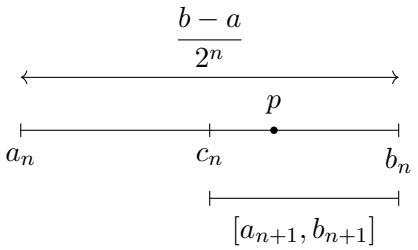
Suppose $f \in C[a, b]$ and $f(a)f(b) < 0$. The intermediate value theorem gives at least one root in (a, b). Set $a_0 = a$, $b_0 = b$. At stage n, compute

$$c_n = \frac{a_n + b_n}{2}.$$

If $f(c_n) = 0$, stop. Otherwise retain the half interval on which the endpoint values have opposite signs:

$$(a_{n+1}, b_{n+1}) = \begin{cases} (a_n, c_n), & f(a_n)f(c_n) < 0, \\ (c_n, b_n), & f(c_n)f(b_n) < 0. \end{cases}$$

This is also called the binary search method.



The diagram shows the retained right half when $f(c_n)f(b_n) < 0$.

Proposition 3.2. *If bisection does not terminate at an exact root, its nested intervals contain a common point p with $f(p) = 0$. Moreover,*

$$b_n - a_n = \frac{b-a}{2^n}, \qquad |c_n - p| \le \frac{b_n - a_n}{2} = \frac{b-a}{2^{n+1}}.$$

Proof. Each retained interval has half the length of the preceding interval, and its endpoint values have opposite signs. The nested intervals have length tending to zero, so they have exactly one common point p. Both endpoint sequences tend to p. Continuity and the opposite endpoint signs imply $f(p) = 0$. Since $p \in [a_n, b_n]$, its distance from the midpoint is at most half the interval length.

To guarantee $|c_n - p| \le \varepsilon$, it suffices to stop when $(b_n - a_n)/2 \le \varepsilon$. For the strict error criterion $|c_n - p| < \varepsilon$, choose an integer $n \ge 0$ such that

$$\frac{b-a}{2^{n+1}} < \varepsilon, \qquad n > \frac{\log(b-a) - \log(2\varepsilon)}{\log 2}.$$

Here n counts interval halvings before forming c_n; the computed midpoints are $c_0, \ldots, c_n$.

Example 3.3. Let $f(x) = x^3 + x - 1$. Since $f(0) = -1$ and $f(1) = 1$, bisection applies on $[0, 1]$. Also $f'(x) = 3x^2 + 1 > 0$, so the root is unique. The first computations are

n	a_n	c_n	b_n	sign of $f(c_n)$
0	0	0.5	1	$-$
1	0.5	0.75	1	$+$
2	0.5	0.625	0.75	$-$
3	0.625	0.6875	0.75	$+$
4	0.625	0.65625	0.6875	$-$
5	0.65625	0.671875	0.6875	$-$
6	0.671875	0.6796875	0.6875	$-$
7	0.6796875	0.68359375	0.6875	$+$
8	0.6796875	0.681640625	0.68359375	$-$
9	0.681640625	0.6826171875	0.68359375	$+$
10	0.681640625	0.68212890625	0.6826171875	$-$

For example, $f(0.625) = 0.244140625 + 0.625 - 1 = -0.130859375$, whereas $f(0.6875) = 0.324951171875 + 0.6875 - 1 = 0.012451171875$. At stage ten,

$$c_{10} = 0.68212890625, \qquad |c_{10} - p| \le 2^{-11} = 0.00048828125 < 0.5 \times 10^{-3}.$$

Thus p lies within 0.0005 of c_{10}. If the displayed approximation is rounded to 0.6821, the rounding must also be included in its error bound. One further bisection step gives $c_{11} = 0.682373046875$ with $f(c_{11}) > 0$, so

$$0.68212890625 < p < 0.682373046875.$$

This bracket verifies that the root rounds to 0.682 to three decimal places.

Exercise 3.1. For a starting interval of length one, determine how many bisection halvings suffice to make the midpoint error less than 0.5×10^{-6}.

Solution. The estimate is $e_n \le 2^{-(n+1)}$. Hence it suffices that

$$2^{-(n+1)} < 0.5 \times 10^{-6} \quad \Longleftrightarrow \quad 2^{-n} < 10^{-6} \quad \Longleftrightarrow \quad n > \frac{6}{\log_{10} 2} \approx 19.9316.$$

Thus $n = 20$ halvings suffice, followed by evaluation of the midpoint c_{20}. With the convention that the original midpoint is step one, this is the twenty-first midpoint.

Note 3.4. Bisection guarantees convergence and supplies an explicit error bound, but it is slow. Its errors satisfy a geometric bound $e_n \le A(1/2)^n$, where $A = (b-a)/2$. The individual ratios e_{n+1}/e_n need not be bounded by $1/2$ or have a limit. The estimate above is an exact-arithmetic estimate; it does not include roundoff in midpoint calculations or function evaluations. In floating-point arithmetic a computed midpoint may eventually coincide with an endpoint.

3.3 Fixed points and equivalent equations

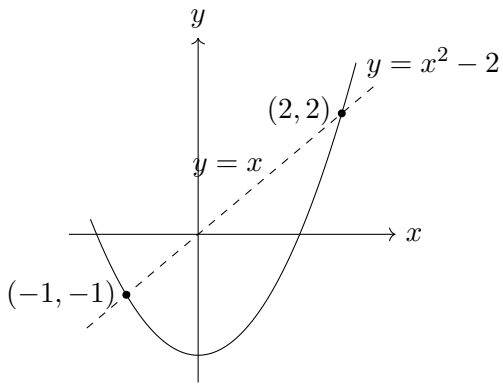
Definition 3.5. A point p is a fixed point of a function g if $g(p) = p$. Given an initial approximation p_0, fixed-point iteration is

$$p_{n+1} = g(p_n), \qquad n \ge 0.$$

For example, $g(x) = x^3$ has the three fixed points $-1, 0, 1$. For $g(x) = x^2 - 2$, the fixed-point equation is

$$x^2 - 2 = x \quad \Longleftrightarrow \quad (x-2)(x+1) = 0,$$

so its fixed points are 2 and -1. Geometrically, fixed points are intersections of $y = g(x)$ with $y = x$.



If $f(p) = 0$, then p is a fixed point of $g(x) = x - f(x)$ and also of $g(x) = x - 2f(x)$. Conversely, if $g(p) = p$, then $f(x) = x - g(x)$ has a root at p. Different equivalent fixed-point equations can lead to very different iterations.

If a well-defined sequence $p_{n+1} = g(p_n)$ converges to a point p in the domain of g, and g is continuous at p, then

$$g(p) = g\left(\lim_{n\to\infty} p_n\right) = \lim_{n\to\infty} g(p_n) = \lim_{n\to\infty} p_{n+1} = p.$$

This identifies a limit as a fixed point, but it does not establish that the sequence converges.

Exercise 3.2. Show that a continuous function $g : [a, b] \to [a, b]$ has a fixed point. If g is differentiable on (a, b) and $|g'(x)| \le k < 1$ there, show that its fixed point is unique.

Solution. Let $h(x) = g(x) - x$. Since $g(a) \ge a$ and $g(b) \le b$, we have $h(a) \ge 0$ and $h(b) \le 0$. By continuity, $h(p) = 0$ for some $p \in [a, b]$. If p and q were distinct fixed points, the mean value theorem would give

$$|p - q| = |g(p) - g(q)| \le k\,|p - q|\,,$$

which is impossible for $k < 1$ unless $p = q$.

Definition 3.6. A function g is a contraction on $[a, b]$ if there is a constant $0 \le k < 1$ such that

$$|g(x) - g(y)| \le k\,|x - y| \qquad (x, y \in [a, b]).$$

In particular, continuity on $[a, b]$ and a derivative bound $|g'| \le k < 1$ on (a, b) imply this inequality.

Theorem 3.7 (Fixed-point iteration)**.** *Suppose $g : [a, b] \to [a, b]$ is a contraction with constant $k < 1$. Then g has a unique fixed point $p \in [a, b]$, and for every $p_0 \in [a, b]$, the iterates $p_{n+1} = g(p_n)$ converge to p. For $n \ge 0$,*

$$|p_n - p| \le k^n\,|p_0 - p| \le k^n \max\{p_0 - a, b - p_0\}.$$

For $0 < k < 1$, there are also the computable estimates

$$|p - p_n| \le \frac{k^n}{1 - k}\,|p_1 - p_0| \qquad (n \ge 0),$$

and

$$|p - p_n| \le \frac{k}{1 - k}\,|p_n - p_{n-1}| \qquad (n \ge 1).$$

Proof. The existence and uniqueness argument just given applies because a contraction is continuous; uniqueness follows directly from its defining inequality. The self-mapping assumption keeps all iterates in $[a, b]$. Since $g(p) = p$,

$$|p_n - p| \le k\,|p_{n-1} - p| \le \cdots \le k^n\,|p_0 - p| \longrightarrow 0.$$

Both p and p_0 lie in $[a, b]$, giving the first stated bound. Also,

$$|p_{j+1} - p_j| \le k\,|p_j - p_{j-1}| \le k^j\,|p_1 - p_0|\,.$$

For $m > n$, telescope and use the geometric series:

$$\begin{aligned} |p_m - p_n| &\le \sum_{j=n}^{m-1} |p_{j+1} - p_j| \\ &\le k^n |p_1 - p_0| (1 + k + \cdots + k^{m-n-1}) \\ &\le \frac{k^n}{1-k} |p_1 - p_0| . \end{aligned}$$

Letting $m \to \infty$ gives the second bound. Finally,

$$|p - p_n| \le \sum_{j=n}^{\infty} |p_{j+1} - p_j| \le \sum_{r=1}^{\infty} k^r |p_n - p_{n-1}| = \frac{k}{1-k} |p_n - p_{n-1}| .$$

If $k = 0$, g is constant and the first iterate is the fixed point.

The bounds become more effective as k approaches zero and less effective as k approaches one. The constant k is an upper bound; it need not equal the asymptotic error constant.

Proposition 3.8. *Suppose g is continuously differentiable near a fixed point p, and $p_n \to p$ without reaching p in finitely many steps. Then*

$$\lim_{n\to\infty} \frac{|p_{n+1} - p|}{|p_n - p|} = |g'(p)| .$$

Thus $0 < |g'(p)| < 1$ gives linear convergence, while $g'(p) = 0$ gives superlinear convergence.

Proof. By the mean value theorem, there is ξ_n between p_n and p such that

$$\frac{|p_{n+1} - p|}{|p_n - p|} = \frac{|g(p_n) - g(p)|}{|p_n - p|} = |g'(\xi_n)| .$$

Since $\xi_n \to p$, continuity of g' gives the limit.

Example 3.9. Let $g(x) = (x^2 - 1)/3$ on $[-1, 1]$. Since $0 \le x^2 \le 1$,

$$-\frac{1}{3} \le g(x) \le 0, \qquad |g'(x)| = \frac{2|x|}{3} \le \frac{2}{3}.$$

Thus g maps $[-1, 1]$ into itself and is a contraction. Its unique fixed point in this interval satisfies

$$\frac{p^2 - 1}{3} = p, \qquad p^2 - 3p - 1 = 0, \qquad p = \frac{3 - \sqrt{13}}{2}.$$

The other algebraic solution $(3 + \sqrt{13})/2$ lies outside the interval.

Exercise 3.3. Consider the fixed-point equation $p = 3^{-p}$. Find an interval on which iteration converges and give a computable error bound.

Solution. Take $I = [1/3, 1]$ and $g(x) = 3^{-x}$. On I, g is decreasing and

$$\frac{1}{3} = g(1) \le g(x) \le g(1/3) = 3^{-1/3} < 1.$$

Moreover,

$$|g'(x)| = (\log 3)3^{-x} \le k = (\log 3)3^{-1/3} < 1.$$

For instance, $\log 3 < 1.1$ and $3^{-1/3} < 0.7$ give $k < 0.77$. Hence every initial point in I converges to the unique fixed point there. For $p_0 = 1$, $p_1 = 1/3$, so

$$|p_n - p| \le \frac{2k^n}{3(1-k)}.$$

This provides an approximation and an error bound without requiring an elementary formula for p.

Exercise 3.4. For $f(x) = 4x^2 + x$, find the fixed points of $g_1(x) = x - f(x)$ and $g_2(x) = x - 2f(x)$, and compare the derivative tests at their fixed points.

Solution. Both maps have fixed points exactly at the roots $x(4x + 1) = 0$, namely 0 and $-1/4$. Explicitly,

$$g_1(x) = -4x^2, \quad g_1'(x) = -8x, \qquad g_2(x) = -x - 8x^2, \quad g_2'(x) = -1 - 16x.$$

At zero, $g_1'(0) = 0$. For example, on $[-1/16, 1/16]$, g_1 maps into $[-1/64, 0]$ and $|g_1'| \le 1/2$, so iteration converges to zero there. Also $|p_{n+1}| = 4\,|p_n|^2$ for this iteration. At $-1/4$, $g_1'(-1/4) = 2$, so this fixed point is locally repelling. For g_2, the derivatives at 0 and $-1/4$ are -1 and 3, respectively. The contraction derivative test gives no conclusion at zero, and $-1/4$ is locally repelling. The condition $|g'(p)| = 1$ requires further analysis; it does not establish either convergence or divergence.

Example 3.10. Consider again $f(x) = x^3 + x - 1$, whose root is $p \approx 0.682327803828$. Three equivalent fixed-point maps are

$$g_1(x) = 1 - x^3, \qquad g_2(x) = \sqrt[3]{1 - x}, \qquad g_3(x) = \frac{1 + 2x^3}{1 + 3x^2}.$$

For g_3, the fixed-point equation gives

$$3x^3 + x = 2x^3 + 1 \quad \Longleftrightarrow \quad x^3 + x - 1 = 0.$$

At the root,

$$|g_1'(p)| = 3p^2 \approx 1.396713696 > 1.$$

By continuity of g_1', distances from p increase for distinct points sufficiently close to p. Thus p is repelling for this iteration; nearby starting values do not generally converge to it. For the second map,

$$g_2'(x) = -\frac{1}{3}(1 - x)^{-2/3}, \qquad |g_2'(p)| = \frac{1}{3p^2} \approx 0.715966345 < 1.$$

Continuity of g_2' near p provides a smaller invariant interval on which it is a contraction, so sufficiently close starting values converge linearly. This is a local assertion: the derivative is unbounded near $x = 1$. Finally,

$$g_3'(x) = \frac{6x^2(1 + 3x^2) - 6x(1 + 2x^3)}{(1 + 3x^2)^2} = \frac{6x(x^3 + x - 1)}{(1 + 3x^2)^2},$$

so $g_3'(p) = 0$. This suggests faster convergence. In fact, g_3 is the Newton map for f.

3.4 Newton's method

Suppose $f \in C^2$ on an open interval containing a root p, and p_n is an approximation in that interval with $f'(p_n) \neq 0$. Taylor's theorem gives

$$0 = f(p) = f(p_n) + f'(p_n)(p - p_n) + \frac{f''(\xi_n)}{2}(p - p_n)^2,$$

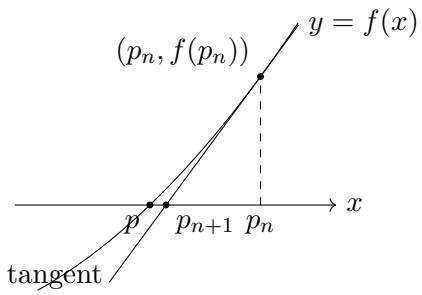
where ξ_n lies between p_n and p. Neglecting the quadratic remainder and solving for p gives the next approximation

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}, \qquad n \geq 0.$$

Geometrically, p_{n+1} is the horizontal intercept of the tangent line

$$y = f(p_n) + f'(p_n)(x - p_n).$$

For $f(x) = x^2 - 2$ and $p_n = 2$, the tangent is $y = 4x - 6$ and its horizontal intercept is $p_{n+1} = 3/2$.



The iteration is defined only while the required derivatives are nonzero; if an exact root is reached, we stop.

Example 3.11. For $f(x) = x^3 + x - 1$, $f'(x) = 3x^2 + 1$, and Newton's formula simplifies to

$$p_{n+1} = p_n - \frac{p_n^3 + p_n - 1}{3p_n^2 + 1} = \frac{2p_n^3 + 1}{3p_n^2 + 1}.$$

Thus the third fixed-point map above is obtained by using the tangent line approximation.

Example 3.12. To approximate $\sqrt{2}$, take $f(x) = x^2 - 2$ on $[1, 2]$. Then

$$p_{n+1} = p_n - \frac{p_n^2 - 2}{2p_n} = \frac{1}{2}\left(p_n + \frac{2}{p_n}\right).$$

Starting with $p_0 = 1$ gives

$$\begin{aligned}
p_1 &= \frac{1}{2}(1 + 2) = \frac{3}{2} = 1.5, \\
p_2 &= \frac{1}{2}\left(\frac{3}{2} + \frac{4}{3}\right) = \frac{17}{12} \approx 1.4166666667, \\
p_3 &= \frac{1}{2}\left(\frac{17}{12} + \frac{24}{17}\right) = \frac{577}{408} \approx 1.4142156863, \\
p_4 &= \frac{665857}{470832} \approx 1.4142135624.
\end{aligned}$$

Theorem 3.13 (Local convergence at a simple root)**.** *Suppose $f \in C^2$ on an open interval containing p, with $f(p) = 0$ and $f'(p) \neq 0$. There exists $\delta > 0$ such that every starting point in $[p - \delta, p + \delta]$ generates Newton iterates that stay in this interval and converge to p, unless an exact root is reached first.*

Proof. Continuity of f' gives a neighborhood of p on which f' is nonzero. There the Newton map

$$g(x) = x - \frac{f(x)}{f'(x)}$$

is continuously differentiable, with

$$g'(x) = 1 - \frac{[f'(x)]^2 - f(x)f''(x)}{[f'(x)]^2} = \frac{f(x)f''(x)}{[f'(x)]^2}.$$

In particular, $g(p) = p$ and $g'(p) = 0$. Fix any $k \in (0, 1)$. By continuity, shrink δ so that $|g'(x)| \leq k$ for $|x - p| \leq \delta$. For such x, the mean value theorem gives

$$|g(x) - p| = |g(x) - g(p)| \leq k\,|x - p| \leq k\delta \leq \delta.$$

Thus g maps the interval into itself and is a contraction there. The fixed-point theorem proves convergence.

Theorem 3.14 (Newton error formula)**.** *Under the preceding hypotheses, suppose the iterates converge to p and do not terminate. For all sufficiently large n, there is a point ξ_n between p_n and p such that*

$$p_{n+1} - p = \frac{f''(\xi_n)}{2f'(p_n)}(p_n - p)^2.$$

Consequently,

$$\lim_{n\to\infty} \frac{|p_{n+1} - p|}{|p_n - p|^2} = \frac{|f''(p)|}{2\,|f'(p)|}.$$

The convergence is at least quadratic. It has exact order two if $f''(p) \neq 0$.

Proof. Rearrange Taylor's formula at p_n:

$$-\frac{f(p_n)}{f'(p_n)} = p - p_n + \frac{f''(\xi_n)}{2f'(p_n)}(p - p_n)^2.$$

Substitution in Newton's formula yields the error identity. Since $p_n \to p$, also $\xi_n \to p$, and continuity gives the limit. Locally $|f'|$ is bounded below by a positive constant, and $|f''|$ is bounded above; hence the identity also supplies a uniform bound $e_{n+1} \leq Ce_n^2$.

For $f(x) = x^2 - 2$ the quadratic error constant is

$$\frac{2}{2(2\sqrt{2})} = \frac{1}{2\sqrt{2}}.$$

For $f(x) = x^3 + x - 1$ it is $3p/(3p^2 + 1)$.

3.5 Multiple roots and modified Newton iteration

Definition 3.15. Suppose m is a positive integer and f is m times differentiable near p. The root p has multiplicity m if

$$f(p) = f'(p) = \cdots = f^{(m-1)}(p) = 0, \qquad f^{(m)}(p) \neq 0.$$

A root of multiplicity one is called simple.

Example 3.16. Let $f(x) = x^2$, whose root is $p = 0$. Newton's method gives, for $p_n \neq 0$,

$$p_{n+1} = p_n - \frac{p_n^2}{2p_n} = \frac{p_n}{2}.$$

Starting with $p_0 = 1$, the iterates are $1, 1/2, 1/4, 1/8, \ldots$. Hence

$$e_{n+1} = \frac{1}{2} e_n.$$

The convergence is linear, even though f is a polynomial. More generally, for $f(x) = x^m$ with an integer $m \geq 1$,

$$p_{n+1} = p_n - \frac{p_n^m}{m p_n^{m-1}} = \frac{m-1}{m} p_n.$$

For $m > 1$, the asymptotic error constant is $(m-1)/m$; for $m = 1$, Newton's method reaches the root in one step.

Exercise 3.5. Suppose $f \in C^{m+1}$ near a root p of multiplicity $m > 1$. Prove that Newton's method converges locally with linear error constant $(m-1)/m$. Show that the modified Newton iteration

$$p_{n+1} = p_n - m \frac{f(p_n)}{f'(p_n)}$$

converges locally with at least quadratic order.

Solution. Taylor's formula with integral remainder gives the local factorization

$$f(x) = (x-p)^m h(x), \qquad h(x) = \frac{1}{(m-1)!} \int_0^1 (1-t)^{m-1} f^{(m)}(p + t(x-p))\, dt.$$

The hypothesis $f \in C^{m+1}$ implies $h \in C^1$, and

$$h(p) = \frac{f^{(m)}(p)}{m!} \neq 0, \qquad h'(p) = \frac{f^{(m+1)}(p)}{(m+1)!}.$$

Write $u = x - p$. Then

$$f'(x) = u^{m-1}\big(m h(x) + u h'(x)\big).$$

The bracketed factor is nonzero near p, so $f'(x) \neq 0$ there when $x \neq p$. For the ordinary Newton map N,

$$N(x) - p = u - \frac{u h(x)}{m h(x) + u h'(x)} = u \frac{(m-1)h(x) + u h'(x)}{m h(x) + u h'(x)}.$$

The last factor tends to $(m-1)/m < 1$. Choose $r \in ((m-1)/m, 1)$ and a sufficiently small $\delta > 0$ so that

$$|N(x) - p| \le r\,|x - p| \qquad (0 < |x - p| \le \delta).$$

The iterates stay in this neighborhood and converge to p; if an iterate equals p, we stop. For a nonterminating sequence, division by $|p_n - p|$ gives

$$\lim_{n\to\infty} \frac{e_{n+1}}{e_n} = \frac{m-1}{m}.$$

For the modified map M, cancellation instead gives

$$M(x) - p = u - m\frac{uh(x)}{mh(x) + uh'(x)} = \frac{u^2 h'(x)}{mh(x) + uh'(x)}.$$

The coefficient of u^2 is bounded near p. Thus $|M(x) - p| \le C\,|x - p|^2$ there. Shrink the neighborhood until $C\delta < 1$; this proves invariance and local convergence. For a nonterminating sequence,

$$\lim_{n\to\infty} \frac{e_{n+1}}{e_n^2} = \left|\frac{h'(p)}{mh(p)}\right| = \left|\frac{f^{(m+1)}(p)}{m(m+1)f^{(m)}(p)}\right|.$$

If this constant is positive, the order is exactly two. If it is zero, the estimate is still at least quadratic, and the convergence may be faster.

Example 3.17. Consider

$$f(x) = \sin x + x^2 \cos x - x^2 - x.$$

We have

$$\begin{aligned} f'(x) &= \cos x + 2x\cos x - x^2 \sin x - 2x - 1, \\ f''(x) &= -\sin x + 2\cos x - 4x \sin x - x^2 \cos x - 2, \\ f'''(x) &= -\cos x - 6\sin x - 6x\cos x + x^2 \sin x. \end{aligned}$$

Therefore

$$f(0) = f'(0) = f''(0) = 0, \qquad f'''(0) = -1 \neq 0,$$

so $p = 0$ has multiplicity three. Ordinary Newton iteration has asymptotic linear error constant $2/3$.

Exercise 3.6. For this function, estimate the number of ordinary Newton iterations needed for an error less than 0.5×10^{-6}, using the asymptotic factor $2/3$ and the initial error $e_0 = 1$. Then compute the modified Newton iterates from $p_0 = 1$ through p_5.

Solution. The asymptotic model is

$$e_n \approx \frac{2}{3} e_{n-1} \approx \left(\frac{2}{3}\right)^n e_0.$$

Setting $e_0 = 1$ and solving the model inequality gives

$$\left(\frac{2}{3}\right)^n < 0.5 \times 10^{-6}, \qquad n > \frac{\log_{10}(0.5) - 6}{\log_{10}(2/3)} \approx 35.7828.$$

Thus the model predicts 36 iterations. This is an asymptotic estimate, not a guaranteed bound from $p_0 = 1$. Direct high-precision iteration gives

$$p_{36} \approx 8.1867087566 \times 10^{-7}, \quad p_{37} \approx 5.4578080718 \times 10^{-7}, \quad p_{38} \approx 3.6385397075 \times 10^{-7};$$

the first iterate meeting the requested tolerance is p_{38}.

The modified formula, using multiplicity three, is

$$p_{n+1} = p_n - \frac{3(\sin p_n + p_n^2 \cos p_n - p_n^2 - p_n)}{\cos p_n + 2p_n \cos p_n - p_n^2 \sin p_n - 2p_n - 1}.$$

For example,

$$p_1 = 1 - \frac{3(\sin 1 + \cos 1 - 2)}{3\cos 1 - \sin 1 - 3} \approx 0.164770719582.$$

Successive substitution gives

n	p_n
0	1
1	0.1647707195822416
2	0.01620733771143968
3	0.0002465414376881191
4	$6.072229811602935 \times 10^{-8}$
5	$3.687196585448568 \times 10^{-15}$

The rapid decrease is consistent with the quadratic error formula. Indeed, expansion near zero gives

$$f(x) = -\frac{x^3}{6} - \frac{x^4}{2} + \frac{x^5}{120} + O(x^6).$$

Thus $h(0) = -1/6$ and $h'(0) = -1/2$, so the modified Newton error constant is

$$\left| \frac{h'(0)}{3h(0)} \right| = 1.$$

Near a multiple root, direct evaluations of f and f' can involve severe cancellation. The displayed iterates use sufficient precision to retain these small values.

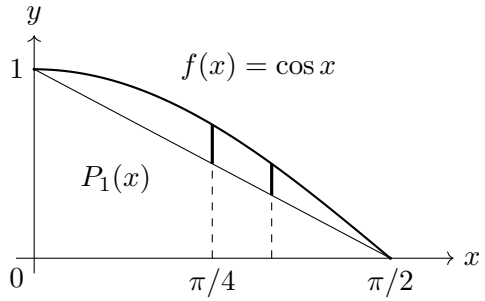
Chapter 4

Polynomial Interpolation

4.1 The interpolation problem

Example 4.1. On $[0, \pi/2]$, approximate $f(x) = \cos x$ by the line through $(0, 1)$ and $(\pi/2, 0)$:

$$P_1(x) = 1 - \frac{2x}{\pi}.$$



The values agree at the endpoints, but need not agree between them. In particular,

$$P_1(\pi/4) = \frac{1}{2}, \qquad f(\pi/4) = \frac{\sqrt{2}}{2}, \qquad |P_1(\pi/4) - f(\pi/4)| = \frac{\sqrt{2}-1}{2} \approx 0.2071.$$

Similarly,

$$P_1(\pi/3) = \frac{1}{3}, \qquad f(\pi/3) = \frac{1}{2}, \qquad |P_1(\pi/3) - f(\pi/3)| = \frac{1}{6} \approx 0.1667.$$

There are higher-degree polynomials through the same two points. For example,

$$P_2(x) = 1 - \frac{4x^2}{\pi^2}, \qquad P_3(x) = 1 - \frac{2x}{3\pi} - \frac{4x^2}{\pi^2} + \frac{8x^3}{3\pi^3}.$$

Indeed, $P_2(0) = P_3(0) = 1$, while $P_2(\pi/2) = 1 - 1 = 0$ and $P_3(\pi/2) = 1 - 1/3 - 1 + 1/3 = 0$. More generally, every polynomial of the form

$$P_1(x) + x(x - \pi/2)q(x)$$

has these endpoint values. Thus a degree restriction is essential for uniqueness.

Definition 4.2. Given pairwise distinct nodes $x_0, \dots, x_n \in \mathbb{R}$ and values $y_0, \dots, y_n \in \mathbb{R}$, an interpolating polynomial is a polynomial P such that $P(x_i) = y_i$ for $0 \le i \le n$.

Theorem 4.3. *For pairwise distinct nodes $x_0, \dots, x_n$, there is exactly one interpolating polynomial of degree at most n.*

Proof. Define the Lagrange cardinal polynomials by

$$L_i(x) = \prod_{\substack{0 \le j \le n \\ j \ne i}} \frac{x - x_j}{x_i - x_j}, \qquad 0 \le i \le n.$$

For $n = 0$, the empty product is 1. Distinctness of the nodes makes every denominator nonzero, and

$$L_i(x_j) = \begin{cases} 1, & j = i, \\ 0, & j \ne i. \end{cases}$$

Consequently,

$$P(x) = \sum_{i=0}^{n} y_i L_i(x)$$

has degree at most n and satisfies $P(x_j) = y_j$. If Q is another such polynomial, $H = P - Q$ has degree at most n and has the $n + 1$ distinct roots $x_0, \dots, x_n$. Hence $H = 0$, so $P = Q$.

4.2 Lagrange and Newton forms

Example 4.4. For two nodes,

$$L_0(x) = \frac{x - x_1}{x_0 - x_1}, \qquad L_1(x) = \frac{x - x_0}{x_1 - x_0}, \qquad P(x) = y_0 L_0(x) + y_1 L_1(x).$$

At x_0, this gives $y_0 \cdot 1 + y_1 \cdot 0 = y_0$; at x_1, it gives y_1. For three nodes,

$$\begin{aligned} L_0(x) &= \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}, \\ L_1(x) &= \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}, \\ L_2(x) &= \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}. \end{aligned}$$

Each polynomial is 1 at its own node and 0 at the other two nodes, so $P = y_0 L_0 + y_1 L_1 + y_2 L_2$ interpolates all three values.

Definition 4.5. For data $f(x_i)$ at distinct nodes, define the divided differences recursively:

$$f[x_i] = f(x_i), \qquad f[x_i, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i} \quad (k \ge 1).$$

In particular, $f[x_i, x_{i+1}] = (f(x_{i+1}) - f(x_i))/(x_{i+1} - x_i)$. These are numbers determined by the data.

Exercise 4.1. Prove that the interpolating polynomial can be written in Newton form:

$$\begin{aligned} P_n(x) = {} & f[x_0] + f[x_0, x_1](x - x_0) \\ & + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \cdots \\ & + f[x_0, \ldots, x_n] \prod_{j=0}^{n-1} (x - x_j). \end{aligned}$$

Solution. For a list of $k+1$ nodes, let $c[x_i, \ldots, x_{i+k}]$ be the coefficient of x^k in its interpolating polynomial. For one node, this is $f(x_i)$. If A interpolates $x_i, \ldots, x_{i+k-1}$ and B interpolates $x_{i+1}, \ldots, x_{i+k}$, then

$$R(x) = \frac{(x - x_i)B(x) - (x - x_{i+k})A(x)}{x_{i+k} - x_i}$$

has the correct value at every node in the combined list. At an interior node, both A and B equal the prescribed value, and at either endpoint one term vanishes. Thus uniqueness identifies R as the interpolating polynomial. Its leading coefficient gives

$$c[x_i, \ldots, x_{i+k}] = \frac{c[x_{i+1}, \ldots, x_{i+k}] - c[x_i, \ldots, x_{i+k-1}]}{x_{i+k} - x_i}.$$

Induction identifies c with the divided difference. Now $P_k - P_{k-1}$ vanishes at $x_0, \ldots, x_{k-1}$ and has degree at most k. It therefore equals a constant times $\prod_{j=0}^{k-1}(x - x_j)$. Its leading coefficient is $f[x_0, \ldots, x_k]$. Starting with $P_0 = f[x_0]$ and summing these differences proves the formula.

4.3 Interpolation error

Exercise 4.2. Suppose $f \in C^{n+1}([a, b])$ and $x_0, \ldots, x_n \in [a, b]$ are distinct. Let P_n interpolate f at these nodes. Prove that, for each $x \in [a, b]$ different from the nodes, there is a point $\xi_x \in (a, b)$ such that

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \prod_{i=0}^{n} (x - x_i).$$

At a node the error is zero. Deduce a uniform error bound.

Solution. Write $\omega(t) = \prod_{i=0}^{n}(t - x_i)$ and fix a nonnode x. Set $C = (f(x) - P_n(x))/\omega(x)$ and $G(t) = f(t) - P_n(t) - C\omega(t)$. The function G vanishes at the $n+2$ distinct points $x, x_0, \ldots, x_n$. Repeated Rolle's theorem gives a point ξ_x between their smallest and largest values with $G^{(n+1)}(\xi_x) = 0$. Since $P_n^{(n+1)} = 0$ and $\omega^{(n+1)} = (n+1)!$, $C = f^{(n+1)}(\xi_x)/(n+1)!$. This proves the identity and gives

$$\max_{a \le x \le b} |f(x) - P_n(x)| \le \frac{\max_{a \le t \le b} \left| f^{(n+1)}(t) \right|}{(n+1)!} \max_{a \le x \le b} |\omega(x)| .$$

Chapter 5

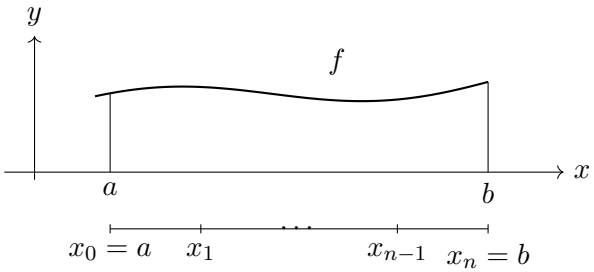
Numerical Integration

5.1 Interpolatory quadrature

Recall that, if $F' = f$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Finding or evaluating F may be inconvenient. Instead, approximate f by its interpolating polynomial and integrate that polynomial.



For distinct nodes $x_0, \ldots, x_n \in [a, b]$, define

$$C_i = \int_a^b L_i(x)\,dx, \qquad Q_n(f) = \sum_{i=0}^{n} C_i f(x_i).$$

Then

$$\int_a^b f(x)\,dx = Q_n(f) + E_n(f), \qquad E_n(f) = \int_a^b \big(f(x) - P_n(x)\big)\,dx.$$

Under the interpolation-error hypotheses, the integrand has the pointwise form

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \prod_{i=0}^{n} (x - x_i).$$

Note that ξ_x can depend on x; it cannot simply be treated as a constant when integrating. Sign information and continuity will be used below.

Exercise 5.1 (Mean value theorem for integrals)**.** Suppose g is Riemann integrable on $[a, b]$, does not change sign, and $f \in C([a, b])$. Prove that there is $c \in [a, b]$ such that

$$\int_a^b f(x)g(x)\,dx = f(c)\int_a^b g(x)\,dx.$$

Solution. First suppose $g \geq 0$. Let $m = \min f$ and $M = \max f$. Then

$$m\int_a^b g \leq \int_a^b fg \leq M\int_a^b g.$$

If $\int g > 0$, the quotient $(\int fg)/(\int g)$ lies between m and M, and the intermediate value theorem supplies c. If $\int g = 0$, the inequalities imply $\int fg = 0$, so any c works. For $g \leq 0$, apply the same argument to $-g$.

5.2 Midpoint and trapezoidal rules

Suppose $a < b$, and put $m = (a + b)/2$. The constant interpolant at the midpoint is $P_0(x) = f(m)$, giving the midpoint rule

$$M(f) = (b - a)f(m).$$

Any line $P_1(x) = f(m) + s(x - m)$ has the same integral, because $\int_a^b (x - m)\,dx = 0$. In particular, adding a linear interpolation term through another node does not change this quadrature value.

Exercise 5.2. For $f \in C^2([a, b])$, prove the midpoint error formula

$$\int_a^b f(x)\,dx = M(f) + \frac{(b - a)^3}{24}f''(\xi_M)$$

for some $\xi_M \in [a, b]$.

Solution. Taylor's theorem about m gives

$$f(x) = f(m) + f'(m)(x - m) + \frac{1}{2}f''(\eta_x)(x - m)^2.$$

Let $\mu = \min_{[a,b]} f''$ and $\nu = \max_{[a,b]} f''$. Since $(x - m)^2 \geq 0$, the remainder lies between $\mu(x - m)^2/2$ and $\nu(x - m)^2/2$. Integrate these inequalities and use cancellation of the linear term:

$$\frac{\mu}{2}\int_a^b (x - m)^2\,dx \leq \int_a^b f - M(f) \leq \frac{\nu}{2}\int_a^b (x - m)^2\,dx.$$

Writing $h = (b - a)/2$,

$$\frac{1}{2}\int_a^b (x - m)^2\,dx = \frac{1}{2}\int_{-h}^h t^2\,dt = \frac{h^3}{3} = \frac{(b - a)^3}{24}.$$

Continuity of f'' now gives ξ_M with the stated error.

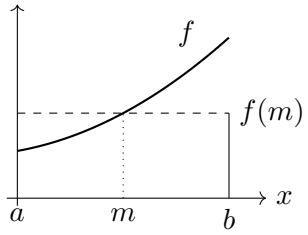
The line interpolating the endpoints is

$$P_1(x) = f[a] + f[a,b](x-a) = f(a) + \frac{f(b)-f(a)}{b-a}(x-a).$$

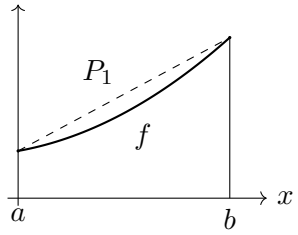
Its integral is

$$\begin{aligned}\int_a^b P_1(x)\,dx &= (b-a)f(a) + \frac{f(b)-f(a)}{b-a}\frac{(b-a)^2}{2} \\ &= \frac{b-a}{2}(f(a)+f(b)).\end{aligned}$$

This is the trapezoidal rule, denoted $T(f)$.



Midpoint rectangle



Endpoint trapezoid

The dashed boundaries belong to the approximating constant or line; the solid curves represent f.

Exercise 5.3. For $f \in C^2([a,b])$, complete the error calculation for the trapezoidal rule:

$$\int_a^b f(x)\,dx = T(f) - \frac{(b-a)^3}{12}f''(\xi_T)$$

for some $\xi_T \in [a,b]$.

Solution. The interpolation remainder is

$$f(x) - P_1(x) = \frac{1}{2}f''(\eta_x)(x-a)(x-b).$$

The product $(x-a)(x-b)$ is nonpositive on $[a,b]$. If $\mu = \min f''$ and $\nu = \max f''$, then

$$\frac{\nu}{2}(x-a)(x-b) \le f(x) - P_1(x) \le \frac{\mu}{2}(x-a)(x-b).$$

With $L = b-a$ and $t = x-a$,

$$\int_a^b (x-a)(x-b)\,dx = \int_0^L (t^2 - Lt)\,dt = \frac{L^3}{3} - \frac{L^3}{2} = -\frac{L^3}{6}.$$

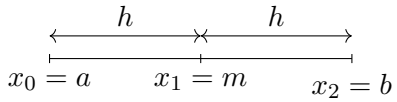
Thus the integrated error is $-L^3/12$ times a number between μ and ν. The intermediate value theorem for f'' proves the formula.

Note 5.1. Both rules are exact for linear polynomials. For a convex function, the midpoint rule underestimates the integral and the trapezoidal rule overestimates it. Their error bounds are, respectively,

$$|E_M| \le \frac{(b-a)^3}{24}\max|f''|, \qquad |E_T| \le \frac{(b-a)^3}{12}\max|f''|.$$

5.3 Simpson's rules

Exercise 5.4 (Simpson's rule)**.** Let $x_0 = a$, $x_1 = m = (a+b)/2$, $x_2 = b$, and $h = (b-a)/2$. Integrate the quadratic interpolant in Newton form to obtain Simpson's rule.



Solution. The interpolant is

$$P_2(x) = f(a) + f[a,m](x-a) + f[a,m,b](x-a)(x-m),$$

where

$$f[a,m] = \frac{f(m)-f(a)}{h}, \qquad f[a,m,b] = \frac{f(b)-2f(m)+f(a)}{2h^2}.$$

Using $t = x - a$, we calculate

$$\int_a^b (x-a)\,dx = 2h^2, \qquad \int_a^b (x-a)(x-m)\,dx = \int_0^{2h} (t^2 - ht)\,dt = \frac{2h^3}{3}.$$

Therefore

$$\begin{aligned} S(f) := \int_a^b P_2(x)\,dx &= 2hf(a) + 2h\big(f(m)-f(a)\big) + \frac{h}{3}\big(f(b)-2f(m)+f(a)\big) \\ &= \frac{h}{3}\big(f(a)+4f(m)+f(b)\big) \\ &= \frac{b-a}{6}\big(f(a)+4f((a+b)/2)+f(b)\big). \end{aligned}$$

Exercise 5.5. For $f \in C^4([a,b])$, prove that

$$\int_a^b f(x)\,dx = S(f) - \frac{(b-a)^5}{2880} f^{(4)}(\xi_S) = S(f) - \frac{h^5}{90} f^{(4)}(\xi_S)$$

for some $\xi_S \in [a,b]$.

Solution. Let P_2 be the quadratic interpolant at a, m, b. Choose a constant c so that

$$H(x) = P_2(x) + c(x-a)(x-m)(x-b)$$

also satisfies $H'(m) = f'(m)$. This is possible because the derivative of $(x-a)(x-m)(x-b)$ at m is $-h^2 \neq 0$. The additional cubic term is odd about m, so $\int_a^b H = \int_a^b P_2 = S(f)$. For any $x \notin \{a,m,b\}$, subtract a suitable multiple of $w(t) = (t-a)(t-m)^2(t-b)$ from $f(t) - H(t)$ to obtain a function vanishing at a, b, x and having a double zero at m. Four applications of Rolle's theorem, counting this double zero, give

$$f(x) - H(x) = \frac{f^{(4)}(\eta_x)}{24}(x-a)(x-m)^2(x-b).$$

The weight is nonpositive. Bound $f^{(4)}(\eta_x)$ between the minimum and maximum of $f^{(4)}$ before integrating. With $t = x - m$,

$$\int_a^b (x-a)(x-m)^2(x-b)\,dx = \int_{-h}^{h} (t^4 - h^2t^2)\,dt = 2h^5\left(\frac{1}{5} - \frac{1}{3}\right) = -\frac{4h^5}{15}.$$

After division by 24, the coefficient is $-h^5/90$. Continuity of $f^{(4)}$ supplies the required ξ_S.

Definition 5.2. A quadrature rule has degree of exactness d if it integrates every polynomial of degree at most d exactly, but fails for at least one polynomial of degree $d+1$.

Example 5.3. Midpoint and trapezoidal quadrature have degree of exactness 1. Their error formulas vanish for linear polynomials, but for $f(x) = x^2$ the respective errors are $(b-a)^3/12$ and $-(b-a)^3/6$. Simpson's rule has degree of exactness 3: its error vanishes for every cubic, while for $f(x) = x^4$ it is $-24(b-a)^5/2880 = -(b-a)^5/120 \neq 0$. Thus the degree of the interpolant and the degree of exactness of its quadrature rule can differ.

Exercise 5.6 (Simpson's 3/8 rule)**.** For four equally spaced nodes $a, a+h, a+2h, a+3h = b$, integrate the cubic interpolant and derive its error for $f \in C^4([a,b])$.

Solution. Set $t = (x-a)/h$, so the nodes become $0, 1, 2, 3$. The cardinal polynomials are

$$\ell_0(t) = -\frac{(t-1)(t-2)(t-3)}{6}, \qquad \ell_1(t) = \frac{t(t-2)(t-3)}{2},$$
$$\ell_2(t) = -\frac{t(t-1)(t-3)}{2}, \qquad \ell_3(t) = \frac{t(t-1)(t-2)}{6}.$$

For example,

$$\int_0^3 \ell_0(t)\,dt = -\frac{1}{6}\left[\frac{t^4}{4} - 2t^3 + \frac{11t^2}{2} - 6t\right]_0^3 = \frac{3}{8},$$

and

$$\int_0^3 \ell_1(t)\,dt = \frac{1}{2}\left[\frac{t^4}{4} - \frac{5t^3}{3} + 3t^2\right]_0^3 = \frac{9}{8}.$$

Reflection about $3/2$ gives the other weights $9/8$ and $3/8$. Hence

$$S_{3/8}(f) = \frac{3h}{8}\big(f(a) + 3f(a+h) + 3f(a+2h) + f(b)\big).$$

To justify the error coefficient, let $g(t) = f(a+ht)$ and write Taylor's formula with integral remainder:

$$g(t) = p_3(t) + \int_0^3 \frac{(t-u)_+^3}{6} g^{(4)}(u)\,du, \qquad (v)_+ = \max\{v, 0\},$$

where p_3 is the Taylor polynomial at 0. The normalized quadrature is exact on p_3. Integrating the remainder and subtracting its quadrature value gives

$$\int_a^b f - S_{3/8}(f) = h^5 \int_0^3 K(u) f^{(4)}(a+hu)\,du,$$

where

$$K(u) = \frac{(3-u)^4}{24} - \frac{3(1-u)_+^3 + 3(2-u)_+^3 + (3-u)^3}{16}.$$

The sign can be checked directly:

$$K(u) = \begin{cases} u^3(2u-3)/48, & 0 \le u \le 1, \\ (u-3/2)^4/24 - 3/128, & 1 \le u \le 2, \\ (3-u)^3(3-2u)/48, & 2 \le u \le 3. \end{cases}$$

All three expressions are nonpositive. To integrate K, apply the normalized error to $g(t) = t^4/24$, whose fourth derivative is 1:

$$\int_0^3 K(u)\,du = \frac{1}{24}\left(\frac{3^5}{5} - \frac{3}{8}(3 + 3\cdot 16 + 81)\right) = -\frac{3}{80}.$$

The mean value theorem for integrals therefore gives

$$\int_a^b f(x)\,dx - S_{3/8}(f) = -\frac{3h^5}{80} f^{(4)}(\xi) = -\frac{(b-a)^5}{6480} f^{(4)}(\xi).$$

This rule has degree of exactness 3 as well.

Example 5.4. Consider

$$I = \int_0^{\pi/4} \sin x\,dx = [-\cos x]_0^{\pi/4} = 1 - \frac{\sqrt{2}}{2} \approx 0.292893218813.$$

With one subinterval, the trapezoidal rule gives

$$T = \frac{\pi}{8}\left(\sin 0 + \sin\frac{\pi}{4}\right) \approx 0.277680183635.$$

With two equal subintervals, Simpson's rule gives

$$S = \frac{\pi}{24}\left(\sin 0 + 4\sin\frac{\pi}{8} + \sin\frac{\pi}{4}\right) \approx 0.292932637840.$$

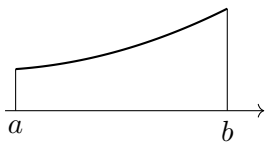
With three equal subintervals, the 3/8 rule gives

$$S_{3/8} = \frac{\pi}{32}\left(\sin 0 + 3\sin\frac{\pi}{12} + 3\sin\frac{\pi}{6} + \sin\frac{\pi}{4}\right) \approx 0.292910702549.$$

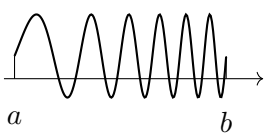
The signed errors $I - Q$ are respectively 0.015213035179, -0.000039419026, and -0.000017483736.

5.4 Composite quadrature

A single low-degree polynomial may approximate a slowly varying function well but miss substantial variation across a long interval. Divide the interval into smaller pieces and apply the rule on each piece.



Slow variation



Oscillation

Exercise 5.7. Let $x_j = a + jh$, $0 \le j \le N$, where $h = (b-a)/N$. Derive composite midpoint and trapezoidal rules, and, for even N, a composite Simpson rule. Give error formulas with their smoothness hypotheses.

Solution. Applying midpoint quadrature on each $[x_j, x_{j+1}]$ and summing gives

$$M_N = h \sum_{j=0}^{N-1} f(x_j + h/2).$$

If $f \in C^2([a,b])$, its error is

$$\int_a^b f - M_N = \frac{h^3}{24} \sum_{j=0}^{N-1} f''(\xi_j) = \frac{(b-a)h^2}{24} f''(\xi_M).$$

Indeed, the average of the N values of f'' lies between its minimum and maximum, so continuity gives the last equality. Summing local trapezoidal rules combines each interior endpoint twice:

$$T_N = h \left(\frac{f(x_0)}{2} + \sum_{j=1}^{N-1} f(x_j) + \frac{f(x_N)}{2} \right).$$

For $f \in C^2([a,b])$, the same averaging argument gives

$$\int_a^b f - T_N = -\frac{h^3}{12} \sum_{j=0}^{N-1} f''(\eta_j) = -\frac{(b-a)h^2}{12} f''(\xi_T).$$

For even N, apply Simpson's rule to each pair $[x_{2j}, x_{2j+2}]$. Each panel has width $2h$:

$$S_N = \frac{h}{3} \left(f(x_0) + 4 \sum_{j=1}^{N/2} f(x_{2j-1}) + 2 \sum_{j=1}^{N/2-1} f(x_{2j}) + f(x_N) \right).$$

If $f \in C^4([a,b])$, there are $N/2$ local errors of the form $-h^5 f^{(4)}(\zeta_j)/90$. Therefore

$$\int_a^b f - S_N = -\frac{h^5}{90} \sum_{j=0}^{N/2-1} f^{(4)}(\zeta_j) = -\frac{(b-a)h^4}{180} f^{(4)}(\xi_S).$$

Taking absolute values gives bounds with the corresponding maximum absolute derivative. For a fixed smooth function and fixed interval, composite midpoint and trapezoidal errors are $O(h^2)$, while composite Simpson error is $O(h^4)$.

5.5 Adaptive Simpson quadrature

For an interval $[a,b]$, write

$$S[a,b] = \frac{b-a}{6}(f(a) + 4f((a+b)/2) + f(b)).$$

An adaptive method subdivides intervals where the local error estimate is too large. It can use smaller intervals in regions where f varies rapidly and larger intervals elsewhere.

Exercise 5.8. Let $m = (a+b)/2$, $S_1 = S[a,b]$, and $S_2 = S[a,m] + S[m,b]$. Derive the usual adaptive Simpson error estimate and describe a recursive algorithm for an absolute tolerance $\varepsilon > 0$. State the limitations of the estimate.

Solution. Write $L = b - a$ and $I = \int_a^b f$. For $f \in C^6$ near the interval, Taylor expansion about m gives

$$I - S_1 = -\frac{L^5}{2880} f^{(4)}(m) + O(L^7).$$

To see the order of the remainder, both the integral and the symmetric quadrature cancel the odd powers in the Taylor polynomial; the rule is exact on the constant and quadratic terms, and its error on the quartic term is $-L^5 f^{(4)}(m)/2880$. The sixth-order remainder contributes $O(L^7)$ after integration or quadrature. For the two half-intervals, the leading errors have total size

$$2\left(\frac{1}{2}\right)^5 = \frac{1}{16}$$

times the leading error on $[a,b]$. More precisely, the fourth derivatives at $m - L/4$ and $m + L/4$ sum to $2f^{(4)}(m) + O(L^2)$, so

$$I - S_2 = -\frac{L^5}{16 \cdot 2880} f^{(4)}(m) + O(L^7).$$

Subtracting the two expansions yields

$$S_2 - S_1 = 15(I - S_2) + O(L^7), \qquad I - S_2 \approx \frac{S_2 - S_1}{15}.$$

The estimated absolute error of S_2 is consequently $|S_2 - S_1|/15$. The corrected value is

$$S_{\text{corr}} = S_2 + \frac{S_2 - S_1}{15},$$

whose local error is $O(L^7)$ under the stated smoothness hypothesis.

For a current interval $[u,v]$ with tolerance τ, perform the following steps:

1. Compute $S_1 = S[u,v]$ and $S_2 = S[u,(u+v)/2] + S[(u+v)/2,v]$. Reuse the endpoint and midpoint values; only the two quarter-point values are new.
2. If $|S_2 - S_1| \le 15\tau$, accept $S_2 + (S_2 - S_1)/15$.
3. Otherwise, apply the same procedure to the two halves, each with tolerance $\tau/2$, and add their returned values.

Start with $[u,v] = [a,b]$ and $\tau = \varepsilon$. The accepted local tolerance budgets sum to at most ε. A numerical implementation must also detect exhausted recursion depth or a midpoint that rounds to an endpoint, and report failure to meet the requested tolerance in that case.

The factor 15 is an asymptotic estimate, not an unconditional error bound. A function can agree with a cubic at all sampled nodes while varying substantially between them. The difference $S_2 - S_1$ may then be misleading. If a known bound $M_4 \ge \max_{[u,v]} \left|f^{(4)}\right|$ is available, the uncorrected refined value satisfies the rigorous bound

$$\left|\int_u^v f - S_2\right| \le 2\frac{((v-u)/2)^5}{2880} M_4 = \frac{(v-u)^5}{46080} M_4.$$

Using this bound as the local acceptance test and returning S_2 gives a certified absolute error bound; summing the accepted bounds proves the global tolerance.

Example 5.5. For the sine integral above, refining the single Simpson panel once gives

$$\begin{aligned} S_2 &= \frac{\pi}{48}\left(\sin 0 + 4\sin\frac{\pi}{16} + 2\sin\frac{\pi}{8} + 4\sin\frac{3\pi}{16} + \sin\frac{\pi}{4}\right) \\ &\approx 0.292895648516. \end{aligned}$$

Since $S_1 \approx 0.292932637840$,

$$\frac{S_2 - S_1}{15} \approx -0.000002465955, \qquad S_{\mathrm{corr}} \approx 0.292893182561.$$

For $\varepsilon = 3 \cdot 10^{-6}$, the estimated error test accepts this refinement. The actual error in S_2 is approximately -0.000002429703; the corrected approximation differs from the exact integral by about $3.63 \cdot 10^{-8}$ in absolute value.

Chapter 6

Numerical Differentiation

6.1 The two-point forward-difference formula

Recall the definition

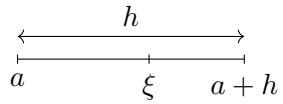
$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

For a small nonzero step h, the two-point forward difference is

$$D_h f(a) = \frac{f(a+h) - f(a)}{h} \approx f'(a).$$

Suppose $h > 0$ and $f \in C^2([a, a+h])$. Taylor's theorem gives

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2} f''(\xi), \qquad a < \xi < a+h.$$



Rearranging,

$$f'(a) = D_h f(a) - \frac{h}{2} f''(\xi), \qquad D_h f(a) - f'(a) = \frac{h}{2} f''(\xi).$$

Thus

$$\left| D_h f(a) - f'(a) \right| \leq \frac{h}{2} \max_{a \leq t \leq a+h} \left| f''(t) \right|.$$

If f is C^2 in a fixed neighborhood of a, this is an $O(h)$ truncation error as $h \to 0^+$. Since $\xi \to a$, its leading expression is $hf''(a)/2$ when $f''(a) \neq 0$.

Example 6.1. Let $f(x) = 1/x$, and approximate $f'(2)$ with $h = 0.1$. Then

$$\begin{aligned} D_{0.1} f(2) &= \frac{1/2.1 - 1/2}{0.1} = \frac{10/21 - 1/2}{1/10} \\ &= -\frac{5}{21} \approx -0.2380952381. \end{aligned}$$

The exact derivative is

$$f'(x) = -\frac{1}{x^2}, \qquad f'(2) = -\frac{1}{4} = -0.25.$$

Therefore the absolute error is

$$\left| -\frac{5}{21} + \frac{1}{4} \right| = \frac{1}{84} \approx 0.0119047619.$$

The Taylor error gives the same scale:

$$f''(x) = \frac{2}{x^3}, \qquad D_{0.1}f(2) - f'(2) = \frac{0.1}{\xi^3}, \qquad 2 < \xi < 2.1.$$

Since $x \mapsto 1/x^3$ decreases on this interval,

$$\frac{0.1}{(2.1)^3} < |D_{0.1}f(2) - f'(2)| < \frac{0.1}{2^3},$$

or numerically $0.0107979700 < 0.0119047619 < 0.0125$. In fact, for every $h > 0$,

$$D_h f(2) = \frac{1/(2+h) - 1/2}{h} = -\frac{1}{2(2+h)}, \qquad D_h f(2) - f'(2) = \frac{h}{4(2+h)}.$$

This displays the first-order dependence on the step directly.

Chapter 7

Initial Value Problems and Euler's Method

7.1 Discrete approximations to an initial value problem

Suppose that the initial value problem

$$y'(t) = f(t, y(t)), \qquad y(t_0) = y_0$$

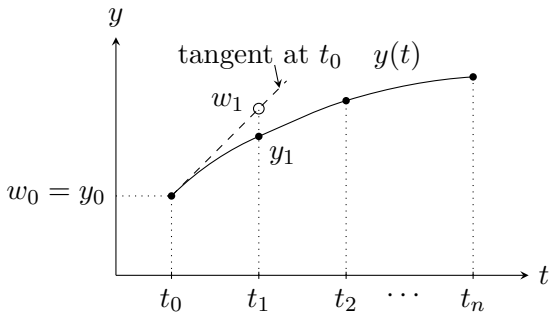
has a unique solution on $[t_0, T]$, where $T > t_0$. Choose a positive integer n and define

$$h = \frac{T - t_0}{n}, \qquad t_i = t_0 + ih, \qquad 0 \le i \le n.$$

Write $y_i = y(t_i)$. A discrete variable method computes numbers w_i approximating y_i, with $w_0 = y_0$. A one-step method has the form

$$w_{i+1} = \Phi(t_i, w_i; h),$$

so the next approximation uses the current approximation and the current mesh point.



The filled points represent exact mesh values $y_i = y(t_i)$. The open point is the first Euler approximation w_1, obtained from the tangent line at (t_0, y_0).

7.2 The tangent-line formula

Recall that the tangent line to the solution at t_0 gives

$$y(t_1) \approx y(t_0) + y'(t_0)(t_1 - t_0) = y_0 + hf(t_0, y_0).$$

Thus we set

$$w_1 = w_0 + hf(t_0, w_0).$$

At the next point, $y'(t_1) = f(t_1, y_1)$ is not known exactly. Replacing y_1 by w_1 gives

$$w_2 = w_1 + hf(t_1, w_1).$$

Repeating this construction gives Euler's method:

$$w_0 = y_0, \qquad w_{i+1} = w_i + hf(t_i, w_i), \qquad 0 \le i < n.$$

If y is twice continuously differentiable, Taylor's theorem makes the single-step approximation precise:

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + \frac{h^2}{2} y''(\xi_i), \qquad t_i < \xi_i < t_{i+1}.$$

In particular, the error in one tangent-line step started from the exact value is at most $h^2 \max_{[t_0,T]} |y''| / 2$.

Chapter 8

Direct Methods for Linear Systems

8.1 Gaussian elimination and back substitution

Suppose that $A \in \mathbb{R}^{n\times n}$ and $b \in \mathbb{R}^n$. Gaussian elimination transforms the augmented matrix $[A \mid b]$ into an upper triangular system $Ux = c$. At step k, provided the current pivot $a_{kk}^{(k)}$ is nonzero, we use

$$m_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}, \qquad R_i \leftarrow R_i - m_{ik} R_k, \qquad i = k+1, \dots, n.$$

The entries below the pivot become zero. If all diagonal entries of U are nonzero, back substitution gives

$$x_n = \frac{c_n}{u_{nn}}, \qquad x_i = \frac{c_i - \sum_{j=i+1}^{n} u_{ij} x_j}{u_{ii}}, \quad i = n-1, \dots, 1.$$

Example 8.1. Consider

$$x_1 + x_2 = 3, \qquad 3x_1 - 4x_2 = 2.$$

Elimination gives

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 3 & -4 & 2 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -7 & -7 \end{array}\right] \xrightarrow{R_2 \leftarrow -R_2/7} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \end{array}\right].$$

Thus $x_2 = 1$ and $x_1 = 3 - x_2 = 2$. Continuing with $R_1 \leftarrow R_1 - R_2$ would give

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right].$$

Back substitution obtains the same solution without reducing the entire coefficient matrix to the identity.

Example 8.2. Solve

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 3, \\ 2x_1 + x_2 - 2x_3 &= 3, \\ -3x_1 + x_2 + x_3 &= -6. \end{aligned}$$

The first elimination step gives

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 1 & -2 & 3 \\ -3 & 1 & 1 & -6 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & -3 \\ 0 & 7 & -2 & 3 \end{array}\right].$$

Now use $R_3 \leftarrow R_3 + \frac{7}{3}R_2$:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & -3 \\ 0 & 0 & -2 & -4 \end{array}\right].$$

Back substitution yields

$$x_3 = 2, \qquad x_2 = 1, \qquad x_1 = 3 - 2x_2 + x_3 = 3.$$

Hence $x = (3, 1, 2)^T$.

8.1.1 Operation counts

Count each addition, subtraction, multiplication, or division as one arithmetic operation. We count the dense algorithm, without omitting operations associated with special zero entries. At step k, one row requires one division to form m_{ik} and $n - k + 1$ multiplication–subtraction pairs to update the remaining coefficients and the right-hand side. The eliminated entry is set to zero rather than recomputed. Thus the cost for each of the $n - k$ rows is

$$2(n - k + 1) + 1.$$

With $j = n - k$, the total elimination cost is

$$\begin{aligned} E_n &= \sum_{k=1}^{n-1} (n-k)\{2(n-k+1)+1\} \\ &= \sum_{j=1}^{n-1} j\{2(j+1)+1\} = 2\sum_{j=1}^{n-1} j^2 + 3\sum_{j=1}^{n-1} j \\ &= \frac{(n-1)n(2n-1)}{3} + \frac{3n(n-1)}{2} \\ &= \frac{n(n-1)(4n+7)}{6} = \frac{2}{3}n^3 + \frac{1}{2}n^2 - \frac{7}{6}n. \end{aligned}$$

In particular, elimination takes approximately $\frac{2}{3}n^3$ operations for large n.

The last equation requires one division. The preceding equation requires one multiplication, one subtraction, and one division. In general, the equation with j already known variables requires $2j + 1$ operations. Therefore back substitution costs

$$B_n = 1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

The exact total under this convention is

$$E_n + B_n = \frac{2}{3}n^3 + \frac{3}{2}n^2 - \frac{7}{6}n.$$

If we retain only the leading cost of elimination, the usual comparison is $\frac{2}{3}n^3 + n^2$: elimination is $O(n^3)$, whereas back substitution is $O(n^2)$.

Example 8.3. Suppose elimination for a dense 500×500 system takes one second. Under the model that elapsed time is proportional to the arithmetic count, using the leading elimination count gives

$$\frac{B_{500}}{(2/3)500^3} = \frac{500^2}{(2/3)500^3} = \frac{3}{1000}.$$

Back substitution then takes approximately 0.003 seconds. This estimate compares arithmetic work at the same assumed operation rate.

8.1.2 Pivoting and roundoff

A zero pivot prevents the division defining m_{ik}. A very small pivot can produce large multipliers and large intermediate entries. Subsequent subtraction can magnify rounding errors; a small term may also be lost when combined with a much larger term, an effect called *swamping*.

In partial pivoting, at step k we choose a row $p \geq k$ for which

$$\left|a_{pk}^{(k)}\right| = \max_{i \geq k} \left|a_{ik}^{(k)}\right|$$

and interchange rows p and k, including their right-hand sides. When the pivot is nonzero, this choice ensures $|m_{ik}| \leq 1$. For a nonsingular square matrix a suitable nonzero pivot always exists at each step. Pivoting controls the multipliers, although the remaining entries may still grow during elimination.

8.2 LU factorization and repeated right-hand sides

Definition 8.4. An *LU factorization* of A is a representation

$$A = LU,$$

where L is lower triangular and U is upper triangular. We take L to have diagonal entries equal to one. If elimination proceeds without row exchanges and without zero pivots, its multipliers m_{ik} form the entries of L below the diagonal, and the final coefficient matrix is U. With row exchanges the corresponding factorization is $PA = LU$, where P is a permutation matrix.

Exercise 8.1. Suppose $A = LU$ is nonsingular. Explain why solving $Ax = b$ reduces to two triangular systems, and determine the cost of solving k systems with the same coefficient matrix.

Solution. Set $y = Ux$. Then $LUx = b$ becomes $Ly = b$. Since L and U are nonsingular, the two steps

$$Ly = b, \qquad Ux = y$$

are equivalent to $Ax = b$. For unit lower triangular L, forward substitution gives

$$y_1 = b_1, \qquad y_i = b_i - \sum_{j=1}^{i-1} l_{ij} y_j, \quad i = 2, \ldots, n.$$

It costs $2\sum_{i=1}^{n}(i-1) = n(n-1)$ operations. Back substitution costs n^2 operations, so each new right-hand side costs $2n^2 - n$ operations after factorization.

The leading cost of forming LU is $\frac{2}{3}n^3$. Thus solving $Ax = b^{(1)}, \ldots, Ax = b^{(k)}$ by reusing the factorization costs approximately

$$\frac{2}{3}n^3 + 2kn^2.$$

The factorization is performed once; only the two triangular solves are repeated. For $PA = LU$, replace each right-hand side by $Pb^{(j)}$.

8.3 Special matrices

8.3.1 Strict diagonal dominance

Definition 8.5. A matrix $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ is *strictly diagonally dominant by rows* if

$$|a_{ii}| > \sum_{\substack{j=1\\ j\neq i}}^{n} |a_{ij}|\,, \qquad i = 1, \ldots, n.$$

Example 8.6. The matrix

$$A = \begin{pmatrix} 7 & 2 & 0 \\ 3 & 5 & -1 \\ 0 & 5 & -6 \end{pmatrix}$$

is strictly diagonally dominant because

$$7 > 2, \qquad 5 > 3 + 1, \qquad 6 > 5.$$

In contrast,

$$B = \begin{pmatrix} 6 & 4 & -3 \\ 4 & -2 & 0 \\ -3 & 0 & 1 \end{pmatrix}$$

is not: its first row would require $6 > 4 + 3$, which is false. Its second and third rows also fail the required inequalities.

Exercise 8.2. Prove that a strictly diagonally dominant matrix is nonsingular and that Gaussian elimination can be performed without row exchanges.

Solution. If $Ax = 0$ with $x \neq 0$, choose p such that $|x_p| = \max_i |x_i| > 0$. The pth equation gives

$$|a_{pp}|\,|x_p| \leq \sum_{j\neq p} |a_{pj}|\,|x_j| \leq \left(\sum_{j\neq p} |a_{pj}|\right) |x_p|\,,$$

contradicting strict diagonal dominance. Thus A is nonsingular.

The first pivot a_{11} is nonzero. After eliminating the first column, the remaining matrix is the Schur complement

$$s_{ij} = a_{ij} - \frac{a_{i1}a_{1j}}{a_{11}}, \qquad i, j = 2, \ldots, n.$$

For each $i \geq 2$,

$$\begin{aligned} |s_{ii}| - \sum_{\substack{j=2\\ j\neq i}}^{n} |s_{ij}| &\geq |a_{ii}| - \sum_{\substack{j=2\\ j\neq i}}^{n} |a_{ij}| - \frac{|a_{i1}|}{|a_{11}|} \sum_{j=2}^{n} |a_{1j}| \\ &\geq |a_{ii}| - \sum_{j\neq i} |a_{ij}| > 0. \end{aligned}$$

Hence the Schur complement is again strictly diagonally dominant. Induction shows that every pivot is nonzero, so row exchanges are unnecessary for carrying out elimination in exact arithmetic.

8.3.2 Symmetric positive definite matrices

Definition 8.7. A real matrix A is *symmetric positive definite* if

$$A^T = A, \qquad x^T A x > 0 \quad \text{for every } x \in \mathbb{R}^n \setminus \{0\}.$$

Exercise 8.3. Prove that a real symmetric matrix is positive definite if and only if all its eigenvalues are positive.

Solution. Recall the spectral theorem: a real symmetric matrix has an orthogonal diagonalization $A = Q\Lambda Q^T$, where $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. If A is positive definite and q_i is a unit eigenvector, then

$$\lambda_i = q_i^T A q_i > 0.$$

Conversely, if every $\lambda_i > 0$, put $z = Q^T x$. For $x \neq 0$, orthogonality gives $z \neq 0$, and

$$x^T A x = z^T \Lambda z = \sum_{i=1}^{n} \lambda_i z_i^2 > 0.$$

Exercise 8.4. Show that a symmetric positive definite matrix satisfies

$$a_{ii} > 0, \qquad a_{ij}^2 < a_{ii} a_{jj} \quad (i \neq j).$$

Solution. Taking $x = e_i$ gives $a_{ii} = e_i^T A e_i > 0$. For $i \neq j$, the nonzero vector $te_i + e_j$ gives

$$a_{ii} t^2 + 2a_{ij} t + a_{jj} > 0 \qquad (t \in \mathbb{R}).$$

At $t = -a_{ij}/a_{ii}$ this becomes $a_{jj} - a_{ij}^2/a_{ii} > 0$, which proves the second assertion.

8.3.3 Band and tridiagonal matrices

Definition 8.8. A matrix is *banded* if its nonzero entries lie in a prescribed band about the main diagonal. More precisely, lower and upper bandwidths p, q mean

$$a_{ij} = 0 \quad \text{if } i - j > p \text{ or } j - i > q.$$

Only the diagonals in that band may contain nonzero entries. A *tridiagonal* matrix has $p = q = 1$.

Example 8.9. The matrices

$$C = \begin{pmatrix} 7 & 2 & 0 \\ 3 & 5 & -1 \\ 0 & -5 & -6 \end{pmatrix}, \qquad T = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

are tridiagonal. The matrix C is strictly diagonally dominant. The matrix T is symmetric, but its two interior rows satisfy equality, rather than strict inequality, in the diagonal dominance condition. Nevertheless,

$$x^T T x = x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + (x_3 - x_4)^2 + x_4^2.$$

This expression is positive for every $x \neq 0$: if it vanishes, its nonnegative terms force $x_1 = x_2 = x_3 = x_4 = 0$. Thus T is positive definite.

Note 8.10. Bandedness alone implies neither positive definiteness nor the absence of zero pivots. For example, $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is tridiagonal and nonsingular but requires a row exchange at the first elimination step. Strict diagonal dominance and symmetric positive definiteness are conditions that permit elimination without row exchanges.

8.4 Cholesky factorization

Exercise 8.5. Suppose $A \in \mathbb{R}^{n\times n}$ is symmetric positive definite. Prove that elimination has positive pivots and yields

$$A = LDL^T = GG^T,$$

where L is unit lower triangular, D is diagonal with positive entries, and $G = LD^{1/2}$ is lower triangular with positive diagonal. The factorization $A = GG^T$ is called the *Cholesky factorization.*

Solution. For $n = 1$ the assertion follows from $a_{11} > 0$. For $n > 1$, write

$$A = \begin{pmatrix} \alpha & r^T \\ r & B \end{pmatrix}, \qquad \alpha > 0.$$

One step of elimination leaves $S = B - rr^T/\alpha$. For every $z \neq 0$, let $w = (-r^T z/\alpha, z^T)^T$. Direct multiplication gives

$$z^T S z = w^T A w > 0.$$

Thus S is symmetric positive definite. Induction proves that all subsequent pivots are positive and that elimination requires no row exchanges.

Let $A = LU$, and extract the diagonal of U:

$$D = \operatorname{diag}(u_{11}, \ldots, u_{nn}), \qquad U = D\widehat{U},$$

where $\widehat{U}$ is unit upper triangular. Since $A = A^T$,

$$LD\widehat{U} = \widehat{U}^T D L^T.$$

The factorization into a unit lower triangular matrix and an upper triangular matrix is unique. Indeed, if $L_1U_1 = L_2U_2$, then $L_2^{-1}L_1 = U_2U_1^{-1}$ is both unit lower triangular and upper triangular, hence is the identity. Applying this to the two factorizations above gives $L = \widehat{U}^T$, and therefore

$$A = LDL^T.$$

Since each $d_i = u_{ii} > 0$, we may define

$$D^{1/2} = \operatorname{diag}(\sqrt{d_1}, \ldots, \sqrt{d_n}).$$

Consequently

$$A = LD^{1/2}D^{1/2}L^T = (LD^{1/2})(LD^{1/2})^T = GG^T.$$

Example 8.11. Consider

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 8 \end{pmatrix}.$$

It is positive definite because

$$x^T A x = (x_1 + 2x_2)^2 + 4x_2^2 > 0 \qquad (x \neq 0).$$

Elimination uses $R_2 \leftarrow R_2 - 2R_1$, giving

$$U = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}.$$

Thus

$$D^{1/2} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad G = LD^{1/2} = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix},$$

and

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} = GG^T.$$

Exercise 8.6. Derive formulas for computing the Cholesky factor directly and determine its operation count.

Solution. Comparing the entries of $A = GG^T$ gives

$$g_{jj} = \sqrt{a_{jj} - \sum_{k=1}^{j-1} g_{jk}^2}, \qquad g_{ij} = \frac{a_{ij} - \sum_{k=1}^{j-1} g_{ik} g_{jk}}{g_{jj}}, \quad i = j+1, \ldots, n.$$

Compute columns $j = 1, \ldots, n$ in order. The positive definiteness argument above ensures that each square-root argument is positive.

For each of the $n - j + 1$ entries in column j, accumulating the displayed difference takes $2(j-1)$ arithmetic operations. There are $n - j$ divisions and one square root. Counting square roots separately, the arithmetic count is

$$\begin{aligned} \sum_{j=1}^{n} \{2(n-j+1)(j-1) + (n-j)\} &= \frac{n(n-1)(n+1)}{3} + \frac{n(n-1)}{2} \\ &= \frac{1}{3}n^3 + \frac{1}{2}n^2 - \frac{5}{6}n. \end{aligned}$$

There are also n square roots. The leading cost is $\frac{1}{3}n^3$, approximately half the leading cost of general LU factorization.

Chapter 9

Vector and Matrix Norms and Conditioning

9.1 Vector norms and distances

Definition 9.1. A *vector norm* on $\mathbb{R}^n$ is a function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ such that, for $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$,

(i) $\|x\| \geq 0$, and $\|x\| = 0$ if and only if $x = 0$;

(ii) $\|\alpha x\| = |\alpha| \, \|x\|$;

(iii) $\|x + y\| \leq \|x\| + \|y\|$.

The last property is the triangle inequality.

The standard examples are

$$\|x\|_1 = \sum_{i=1}^{n} |x_i| \,, \qquad \|x\|_2 = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2}, \qquad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i| \,.$$

The 2-norm is the Euclidean norm; the 1-norm is also called the Manhattan norm. More generally, for $1 \leq p < \infty$,

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}.$$

Exercise 9.1 (Cauchy–Bunyakovsky–Schwarz inequality)**.** Prove that, for $x, y \in \mathbb{R}^n$,

$$\sum_{i=1}^{n} |x_i y_i| \leq \|x\|_2 \, \|y\|_2 \,,$$

and deduce the triangle inequality for the Euclidean norm.

Solution. If $y = 0$ the inequality is immediate. Otherwise put $u_i = |x_i|$, $v_i = |y_i|$, and $t = (\sum_i u_i v_i)/(\sum_i v_i^2)$. Since

$$0 \leq \sum_i (u_i - t v_i)^2 = \sum_i u_i^2 - \frac{(\sum_i u_i v_i)^2}{\sum_i v_i^2},$$

we obtain $(\sum_i |x_i y_i|)^2 \le (\sum_i x_i^2)(\sum_i y_i^2)$. Taking square roots proves the assertion. In particular,

$$\begin{aligned}\|x+y\|_2^2 &= \|x\|_2^2 + 2\sum_i x_i y_i + \|y\|_2^2 \\ &\le \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 = (\|x\|_2 + \|y\|_2)^2.\end{aligned}$$

Both sides are nonnegative, so taking square roots proves the triangle inequality.

Exercise 9.2. Verify that $\|\cdot\|_p$ is a norm for $1 \le p < \infty$, and that $\|\cdot\|_\infty$ is a norm.

Solution. Positivity, definiteness, and absolute homogeneity follow from the formulas. For $p = 1$, summing $|x_i + y_i| \le |x_i| + |y_i|$ proves the triangle inequality. For the infinity norm,

$$\max_i |x_i + y_i| \le \max_i |x_i| + \max_i |y_i| .$$

For $p > 1$, let $q = p/(p-1)$. First, Young's inequality

$$ab \le \frac{a^p}{p} + \frac{b^q}{q} \qquad (a, b \ge 0)$$

follows by minimizing $a^p/p - ab + b^q/q$ as a function of a: its minimum is zero at $a = b^{1/(p-1)}$. Apply this inequality to $|u_i| / \|u\|_p$ and $|v_i| / \|v\|_q$ and sum to obtain Hölder's inequality

$$\sum_i |u_i v_i| \le \|u\|_p \|v\|_q .$$

If either vector is zero this inequality holds directly. Now set $s = x + y$. If $s = 0$ there is nothing to prove; otherwise Hölder's inequality gives

$$\begin{aligned}\|s\|_p^p &\le \sum_i (|x_i| + |y_i|)\,|s_i|^{p-1} \\ &\le (\|x\|_p + \|y\|_p)\left(\sum_i |s_i|^{(p-1)q}\right)^{1/q} \\ &= (\|x\|_p + \|y\|_p)\,\|s\|_p^{p-1} .\end{aligned}$$

Division by $\|s\|_p^{p-1}$ proves the triangle inequality.

Example 9.2. For $x = (1, -1, 2)^T$,

$$\|x\|_\infty = 2, \qquad \|x\|_1 = 1 + 1 + 2 = 4, \qquad \|x\|_2 = \sqrt{1 + 1 + 4} = \sqrt{6}.$$

A norm defines the distance $d(x, y) = \|x - y\|$ between vectors. In particular,

$$d_2(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2\right)^{1/2}, \qquad d_\infty(x, y) = \max_i |x_i - y_i| .$$

Positivity and symmetry follow from the norm axioms, and $\|x - z\| \le \|x - y\| + \|y - z\|$ proves the triangle inequality for this distance.

9.2 Matrix norms and induced norms

Definition 9.3. A *matrix norm* on $\mathbb{R}^{n\times n}$ is a scalar-valued function satisfying positivity and definiteness, absolute homogeneity, the triangle inequality, and submultiplicativity:

$$\|A\| \geq 0, \qquad \|A\| = 0 \iff A = 0,$$
$$\|\alpha A\| = |\alpha| \, \|A\|, \qquad \|A + B\| \leq \|A\| + \|B\|, \qquad \|AB\| \leq \|A\| \, \|B\|.$$

It defines the distance $\|A - B\|$ between matrices.

Exercise 9.3. Given any vector norm on $\mathbb{R}^n$, prove that

$$\|A\| = \max_{\|x\|=1} \|Ax\| = \max_{x\neq 0} \frac{\|Ax\|}{\|x\|}$$

defines a matrix norm. This is the *induced* or *operator norm.* Show also that $\|Ax\| \leq \|A\| \, \|x\|$.

Solution. We first justify the maximum. The triangle inequality gives

$$\|x\| \leq \sum_i |x_i| \, \|e_i\| \leq C \, \|x\|_2, \qquad C = \left(\sum_i \|e_i\|^2 \right)^{1/2}.$$

Also $|\|x\| - \|y\|| \leq \|x - y\| \leq C \, \|x - y\|_2$, so the norm is continuous in Euclidean coordinates. On the compact Euclidean unit sphere it has a positive minimum c. By homogeneity,

$$c \, \|x\|_2 \leq \|x\| \leq C \, \|x\|_2 .$$

Thus $\{x : \|x\| = 1\}$ is compact, and the continuous function $x \mapsto \|Ax\|$ attains a maximum there. Normalizing a nonzero x proves the ratio formula and the inequality $\|Ax\| \leq \|A\| \, \|x\|$.

The maximum is nonnegative. If it is zero, then $Ax = 0$ for every x, so $A = 0$; the converse is immediate. Absolute homogeneity follows from that of the vector norm. For every unit vector x,

$$\|(A + B)x\| \leq \|Ax\| + \|Bx\| \leq \|A\| + \|B\|,$$

which proves the matrix triangle inequality upon maximizing. Finally,

$$\|ABx\| \leq \|A\| \, \|Bx\| \leq \|A\| \, \|B\| \, \|x\|.$$

Maximizing over $\|x\| = 1$ proves submultiplicativity.

In particular,

$$\|A\|_\infty = \max_{\|x\|_\infty = 1} \|Ax\|_\infty, \qquad \|A\|_2 = \max_{\|x\|_2 = 1} \|Ax\|_2 .$$

These definitions involve optimization over a unit sphere. For the infinity norm there is a direct computational formula.

Theorem 9.4. *For $A = (a_{ij}) \in \mathbb{R}^{n\times n}$,*

$$\|A\|_\infty = \max_{1\leq i\leq n} \sum_{j=1}^{n} |a_{ij}| .$$

Thus the induced infinity norm is the maximum absolute row sum.

Proof. If $\|x\|_\infty = 1$, then

$$\begin{aligned}\|Ax\|_\infty &= \max_i \left|\sum_j a_{ij}x_j\right| \\ &\le \max_i \sum_j |a_{ij}|\,|x_j| \le \max_i \sum_j |a_{ij}|\,.\end{aligned}$$

To attain the bound, choose a row p of maximum absolute row sum and set

$$x_j = \begin{cases} 1, & a_{pj} \ge 0, \\ -1, & a_{pj} < 0. \end{cases}$$

Then $\|x\|_\infty = 1$ and

$$(Ax)_p = \sum_j |a_{pj}| = \max_i \sum_j |a_{ij}|\,.$$

This proves equality, including when some or all of the entries are zero.

9.3 Spectral radius and the Euclidean matrix norm

Definition 9.5. The *spectral radius* of $A \in \mathbb{R}^{n\times n}$ is

$$\rho(A) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A\},$$

where eigenvalues may be complex.

Exercise 9.4. Prove that $\rho(A) \le \|A\|$ for every induced matrix norm.

Solution. For a real eigenvalue λ with a real unit eigenvector x,

$$|\lambda| = \|\lambda x\| = \|Ax\| \le \|A\|\,.$$

For completeness, consider a complex eigenvalue and eigenvector $Az = \lambda z$ with $z \neq 0$. The norm equivalence established above implies, for any real matrix M,

$$\|M\|_2 \le \frac{C}{c}\,\|M\|\,.$$

Indeed, $\|Mx\|_2 \le c^{-1}\,\|Mx\| \le c^{-1}\,\|M\|\,\|x\| \le (C/c)\,\|M\|\,\|x\|_2$. For $z = u + iv$, the Euclidean norm on complex vectors satisfies

$$\|Mz\|_2^2 = \|Mu\|_2^2 + \|Mv\|_2^2 \le \|M\|_2^2\,(\|u\|_2^2 + \|v\|_2^2).$$

Apply this with $M = A^m$ to obtain

$$|\lambda|^m \le \|A^m\|_2 \le \frac{C}{c}\,\|A^m\| \le \frac{C}{c}\,\|A\|^m\,.$$

Taking mth roots and letting $m \to \infty$ gives $|\lambda| \le \|A\|$. Maximizing over the eigenvalues proves the assertion.

Exercise 9.5. Let $B = A^T A$. Use an orthogonal diagonalization of B to prove

$$\|A\|_2 = \sqrt{\rho(A^T A)}.$$

Identify the singular values of A.

Solution. We have $B^T = B$ and $x^T B x = \|Ax\|_2^2 \geq 0$, so all eigenvalues of B are nonnegative. Order them as

$$0 \leq \lambda_1 \leq \cdots \leq \lambda_n.$$

By the spectral theorem, there is an orthogonal matrix Q such that

$$Q^T B Q = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

For $\|x\|_2 = 1$, set $y = Q^T x$. Then $\|y\|_2 = 1$ and

$$\|Ax\|_2^2 = x^T B x = \sum_i \lambda_i y_i^2 \leq \lambda_n.$$

Equality holds for $x = Qe_n$. Therefore

$$\|A\|_2^2 = \lambda_n = \rho(A^T A).$$

The nonnegative numbers $\sqrt{\lambda_i}$ are the singular values of A, and the largest singular value equals $\|A\|_2$.

9.4 Residuals, errors, and conditioning

Suppose A is nonsingular, $Ax = b$, and $\widehat{x}$ is an approximate solution. We define the *residual*

$$r = b - A\widehat{x}.$$

The forward error is $x - \widehat{x}$, measured by $\|x - \widehat{x}\|$, and, if $x \neq 0$, its relative size is

$$\frac{\|x - \widehat{x}\|}{\|x\|}.$$

With A held fixed, $\widehat{x}$ solves the perturbed system

$$A\widehat{x} = b - r.$$

Thus the absolute backward error in the right-hand side is $\|r\|$ and, for $b \neq 0$, the relative backward error is $\|r\| \,/\, \|b\|$. These quantities concern perturbations of b alone.

Definition 9.6. For a nonsingular matrix and a chosen induced norm, the *condition number* is

$$\operatorname{cond}(A) = \|A\| \left\|A^{-1}\right\|.$$

In the infinity norm we write $\operatorname{cond}_\infty(A) = \|A\|_\infty \left\|A^{-1}\right\|_\infty$.

Exercise 9.6. Suppose $b \neq 0$. Prove that

$$\frac{\|x - \widehat{x}\|}{\|x\|} \leq \operatorname{cond}(A) \frac{\|r\|}{\|b\|}.$$

Explain the error magnification factor.

Solution. Subtracting $A\widehat{x}$ from $Ax = b$ gives

$$A(x - \widehat{x}) = r, \qquad x - \widehat{x} = A^{-1}r.$$

Consequently $\|x - \widehat{x}\| \leq \left\|A^{-1}\right\| \|r\|$. Also $\|b\| = \|Ax\| \leq \|A\| \|x\|$. Combining these inequalities gives

$$\frac{\|x - \widehat{x}\|}{\|x\|} \leq \left\|A^{-1}\right\| \|r\| \frac{\|A\|}{\|b\|},$$

which is the desired bound. When $r \neq 0$, the error magnification factor is

$$\frac{\|x - \widehat{x}\| \,/\, \|x\|}{\|r\| \,/\, \|b\|} \leq \operatorname{cond}(A).$$

Since $\|I\| = 1$ and $I = AA^{-1}$, submultiplicativity also gives $\operatorname{cond}(A) \geq 1$.

Example 9.7. Consider

$$x_1 + x_2 = 2, \qquad 1.0001x_1 + x_2 = 2.0001.$$

Subtracting the first equation from the second gives $0.0001x_1 = 0.0001$, so the exact solution is $x = (1, 1)^T$. Take

$$A = \begin{pmatrix} 1 & 1 \\ 1.0001 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 2.0001 \end{pmatrix}, \qquad \widehat{x} = \begin{pmatrix} -1 \\ 3.0001 \end{pmatrix}.$$

This is an example in which a small residual can accompany a large forward error.

Exercise 9.7. For the preceding system, compute the residual, relative forward and backward errors in the infinity norm, and condition number. Compare the actual magnification factor with the condition number.

Solution. Direct multiplication gives

$$A\widehat{x} = \begin{pmatrix} 2.0001 \\ 2 \end{pmatrix}, \qquad r = b - A\widehat{x} = \begin{pmatrix} -0.0001 \\ 0.0001 \end{pmatrix}, \qquad x - \widehat{x} = \begin{pmatrix} 2 \\ -2.0001 \end{pmatrix}.$$

Thus

$$\|r\|_\infty = 0.0001, \qquad \|x - \widehat{x}\|_\infty = 2.0001, \qquad \|x\|_\infty = 1, \qquad \|b\|_\infty = 2.0001.$$

The relative backward error is

$$\frac{0.0001}{2.0001} = \frac{1}{20001} \approx 4.99975 \times 10^{-5},$$

whereas the relative forward error is 2.0001. Since $\det A = -0.0001$,

$$A^{-1} = \begin{pmatrix} -10000 & 10000 \\ 10001 & -10000 \end{pmatrix}.$$

The maximum absolute row sums give

$$\|A\|_\infty = 2.0001, \qquad \left\|A^{-1}\right\|_\infty = 20001,$$

and hence

$$\operatorname{cond}_\infty(A) = 2.0001(20001) = 40004.0001.$$

The actual error magnification factor is

$$\frac{2.0001}{1/20001} = 40004.0001,$$

so this approximation attains the condition-number bound. A small residual alone does not ensure a small forward error for an ill-conditioned system.

Chapter 10

Iterative Methods for Linear Systems

10.1 Jacobi iteration

Gaussian elimination is a direct method: in exact arithmetic it produces the solution of a nonsingular linear system after finitely many operations. An iterative method starts with an initial guess and generates a sequence intended to converge to the solution.

Example 10.1. Consider

$$3x + y = 5, \qquad x + 2y = 5.$$

Solving the first equation for x and the second for y gives

$$x = \frac{5 - y}{3}, \qquad y = \frac{5 - x}{2}.$$

The Jacobi iteration uses only values from the preceding step:

$$x_{k+1} = \frac{5 - y_k}{3}, \qquad y_{k+1} = \frac{5 - x_k}{2}.$$

Starting with $(x_0, y_0) = (0, 0)$, we obtain

$$(x_1, y_1) = \left(\frac{5}{3}, \frac{5}{2}\right), \qquad (x_2, y_2) = \left(\frac{5 - 5/2}{3}, \frac{5 - 5/3}{2}\right) = \left(\frac{5}{6}, \frac{5}{3}\right).$$

The exact solution is $(1, 2)$: substituting $y = 5 - 3x$ into the second equation gives $x + 2(5 - 3x) = 5$, hence $x = 1$ and $y = 2$.

Suppose $A \in \mathbb{R}^{n \times n}$ has nonzero diagonal entries. Write

$$A = L + D + U,$$

where L is strictly lower triangular, D is diagonal, and U is strictly upper triangular. Rearranging $Ax = b$ gives

$$Dx = b - (L + U)x, \qquad x = D^{-1}b - D^{-1}(L + U)x.$$

For an initial guess $x^{(0)}$, the Jacobi iteration is therefore

$$x^{(k+1)} = D^{-1}\big(b - (L + U)x^{(k)}\big).$$

Equivalently,

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right), \qquad 1 \leq i \leq n.$$

Recall the strict row diagonal dominance condition introduced for direct methods.

Theorem 10.2. *If A is strictly diagonally dominant by rows, then for every b the system $Ax = b$ has a unique solution, and Jacobi iteration converges to that solution for every initial guess.*

Exercise 10.1. Prove the preceding convergence theorem.

Solution. Strict diagonal dominance implies that A is nonsingular, as proved earlier. Let x^* denote its unique solution and put $e^{(k)} = x^{(k)} - x^*$. Subtracting the fixed-point equations yields

$$e^{(k+1)} = -D^{-1}(L + U)e^{(k)}.$$

Set

$$q = \max_i \frac{\sum_{j \neq i} |a_{ij}|}{|a_{ii}|} < 1.$$

Taking the maximum of the componentwise error bounds gives

$$\left\| e^{(k+1)} \right\|_\infty \leq q \left\| e^{(k)} \right\|_\infty \leq q^{k+1} \left\| e^{(0)} \right\|_\infty \longrightarrow 0.$$

10.2 Gauss–Seidel iteration

Gauss–Seidel iteration uses each newly computed component immediately:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right).$$

In matrix form,

$$(D + L)x^{(k+1)} = b - Ux^{(k)}.$$

Example 10.3. For $3x + y = 5$, $x + 2y = 5$, the iteration becomes

$$x_{k+1} = \frac{5 - y_k}{3}, \qquad y_{k+1} = \frac{5 - x_{k+1}}{2}.$$

Starting at $(0, 0)$ gives

$$(x_1, y_1) = \left(\frac{5}{3}, \frac{5}{3} \right), \qquad (x_2, y_2) = \left(\frac{10}{9}, \frac{35}{18} \right).$$

The second update uses x_{k+1}, rather than x_k.

Theorem 10.4. *If A is strictly diagonally dominant by rows, Gauss–Seidel iteration converges to the unique solution of $Ax = b$ for every initial guess.*

Exercise 10.2. Prove the Gauss–Seidel convergence theorem.

Solution. Nonsingularity follows from the preceding argument. Define

$$\alpha_i = \frac{\sum_{j<i} |a_{ij}|}{|a_{ii}|}, \qquad \beta_i = \frac{\sum_{j>i} |a_{ij}|}{|a_{ii}|}, \qquad q = \max_i(\alpha_i + \beta_i) < 1.$$

Let $E = \left\|e^{(k)}\right\|_\infty$. The error equations imply

$$\left|e_i^{(k+1)}\right| \le \frac{\sum_{j<i} |a_{ij}| \left|e_j^{(k+1)}\right|}{|a_{ii}|} + \beta_i E.$$

For $i = 1$, the bound is $\left|e_1^{(k+1)}\right| \le qE$. Inductively, if all preceding new components are bounded by qE, then

$$\left|e_i^{(k+1)}\right| \le (\alpha_i q + \beta_i)E \le (\alpha_i + \beta_i)E \le qE.$$

Consequently $\left\|e^{(k+1)}\right\|_\infty \le q \left\|e^{(k)}\right\|_\infty$, and iteration proves convergence.

10.3 Successive over-relaxation

Successive over-relaxation (SOR) introduces a parameter ω into the Gauss–Seidel update:

$$x_i^{(k+1)} = (1-\omega)x_i^{(k)} + \frac{\omega}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right).$$

Thus the unrelaxed component correction receives weight ω and the old component receives weight $1 - \omega$. The value $\omega = 1$ gives Gauss–Seidel; $\omega > 1$ is over-relaxation. The purpose is to accelerate convergence, but the choice of ω affects both convergence and its rate. For $\omega > 1$ this is an affine combination with a negative old-value weight, rather than a convex average.

Exercise 10.3. Apply SOR with $\omega = 1.5$ to

$$3x + y = 5, \qquad x + 2y = 5,$$

starting with $(x_0, y_0) = (0, 0)$.

Solution. With $\omega = 3/2$, the component formulas are

$$x_{k+1} = -\frac{1}{2}x_k + \frac{1}{2}(5 - y_k), \qquad y_{k+1} = -\frac{1}{2}y_k + \frac{3}{4}(5 - x_{k+1}).$$

For example,

$$x_1 = \frac{5}{2}, \quad y_1 = \frac{3}{4}\left(5 - \frac{5}{2}\right) = \frac{15}{8},$$

$$x_2 = -\frac{5}{4} + \frac{1}{2}\left(5 - \frac{15}{8}\right) = \frac{5}{16}, \qquad y_2 = -\frac{15}{16} + \frac{3}{4}\left(5 - \frac{5}{16}\right) = \frac{165}{64}.$$

Continuing gives

k	x_k	y_k
0	0	0
1	5/2	15/8
2	5/16	165/64
3	135/128	855/512
4	1165/1024	8445/4096

To verify convergence for this choice, subtract the solution $(1, 2)$. The error iteration matrix is

$$T_\omega = \begin{pmatrix} -1/2 & -1/2 \\ 3/8 & -1/8 \end{pmatrix}, \qquad \det(\lambda I - T_\omega) = \lambda^2 + \frac{5}{8}\lambda + \frac{1}{4}.$$

Its roots $(-5 \pm i\sqrt{39})/16$ are distinct and have modulus $1/2$. Over $\mathbb{C}$, $T_\omega = P\,\mathrm{diag}(\lambda_1, \lambda_2)P^{-1}$, so $T_\omega^k \to 0$. Hence the iterates converge to $(1, 2)$.

Chapter 11

Eigenvalue Algorithms

11.1 Power iteration and the Rayleigh quotient

For a real matrix A and a nonzero real vector x, define its Rayleigh quotient by

$$\mu(x) = \frac{x^T A x}{x^T x}.$$

If $Ax = \lambda x$, then $\mu(x) = \lambda$. Power iteration generates $x^{(k+1)} = Ax^{(k)}$; one may rescale each nonzero iterate to avoid excessive growth or decay, since $\mu(cx) = \mu(x)$ for $c \neq 0$.

Exercise 11.1. Let

$$A = \begin{pmatrix} -5 & 4 \\ -8 & 7 \end{pmatrix}.$$

Find its eigenvalues and eigenvectors. Starting from $x^{(0)} = (1, 0)^T$, perform three steps of power iteration and compute the corresponding Rayleigh quotients. Determine their limit.

Solution. The characteristic polynomial is

$$\det(A - \lambda I) = (-5 - \lambda)(7 - \lambda) + 32 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1).$$

For $\lambda = 3$ and $\lambda = -1$, respectively,

$$A - 3I = \begin{pmatrix} -8 & 4 \\ -8 & 4 \end{pmatrix}, \qquad A + I = \begin{pmatrix} -4 & 4 \\ -8 & 8 \end{pmatrix}.$$

Their nullspaces are spanned by

$$v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Successive products give

$$\begin{aligned} x^{(1)} &= A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ -8 \end{pmatrix}, \\ x^{(2)} &= \begin{pmatrix} 25 - 32 \\ 40 - 56 \end{pmatrix} = \begin{pmatrix} -7 \\ -16 \end{pmatrix}, \\ x^{(3)} &= \begin{pmatrix} 35 - 64 \\ 56 - 112 \end{pmatrix} = \begin{pmatrix} -29 \\ -56 \end{pmatrix}. \end{aligned}$$

Also $Ax^{(3)} = (-79, -160)^T$. Therefore

$$\mu(x^{(1)}) = \frac{(-5)(-7) + (-8)(-16)}{25 + 64} = \frac{163}{89} \approx 1.83146,$$
$$\mu(x^{(2)}) = \frac{(-7)(-29) + (-16)(-56)}{49 + 256} = \frac{1099}{305} \approx 3.60328,$$
$$\mu(x^{(3)}) = \frac{(-29)(-79) + (-56)(-160)}{841 + 3136} = \frac{11251}{3977} \approx 2.82902.$$

To justify the limit, decompose the initial vector:

$$x^{(0)} = -v_1 + 2v_2, \qquad x^{(k)} = -3^k v_1 + 2(-1)^k v_2.$$

Consequently

$$3^{-k} x^{(k)} = -v_1 + 2\left(-\frac{1}{3}\right)^k v_2 \longrightarrow -v_1.$$

By continuity and scale invariance of the Rayleigh quotient, $\mu(x^{(k)}) \to \mu(-v_1) = 3$. The quotients need not approach the limit monotonically, and a finite quotient need not lie between the eigenvalues of this nonsymmetric matrix.

11.2 QR iteration

For an invertible real square matrix, an unshifted QR step factors

$$A_k = Q_k R_k, \qquad Q_k^T Q_k = I,$$

with R_k upper triangular, and sets

$$A_{k+1} = R_k Q_k = Q_k^T A_k Q_k.$$

Thus every step is an orthogonal similarity and preserves eigenvalues. When the iterates approach a diagonal matrix, its diagonal entries give the eigenvalues. This behavior must be justified for the matrix under consideration.

Exercise 11.2. Perform three unshifted QR iterations for

$$A_0 = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix},$$

using positive diagonal entries in R_k. Use the resulting diagonal entries to approximate the eigenvalues.

Solution. Write the columns as $a_1 = (2, 1)^T$, $a_2 = (1, 3)^T$. Gram–Schmidt gives

$$q_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad (q_1^T a_2) q_1 = \frac{5}{5} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Hence

$$u_2 = a_2 - (q_1^T a_2) q_1 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \qquad q_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

It follows that

$$Q_0 = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \qquad R_0 = Q_0^T A_0 = \begin{pmatrix} \sqrt{5} & \sqrt{5} \\ 0 & \sqrt{5} \end{pmatrix},$$

$$A_1 = R_0 Q_0 = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Applying the same procedure to A_1 gives

$$Q_1 = \frac{1}{\sqrt{10}}\begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix}, \qquad R_1 = \begin{pmatrix} \sqrt{10} & 5/\sqrt{10} \\ 0 & 5/\sqrt{10} \end{pmatrix}.$$

Thus

$$A_2 = R_1 Q_1 = \frac{1}{10}\begin{pmatrix} 10 & 5 \\ 0 & 5 \end{pmatrix}\begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 7/2 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}.$$

The third factorization is

$$Q_2 = \frac{1}{\sqrt{50}}\begin{pmatrix} 7 & -1 \\ 1 & 7 \end{pmatrix}, \qquad R_2 = \begin{pmatrix} 5/\sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{2} \end{pmatrix}.$$

For example, the first diagonal entry of R_2 is $\sqrt{(7/2)^2 + (1/2)^2} = 5/\sqrt{2}$, and the other entries follow from $Q_2^T A_2$. Multiplication gives

$$A_3 = R_2 Q_2 = \begin{pmatrix} 18/5 & 1/5 \\ 1/5 & 7/5 \end{pmatrix}.$$

The diagonal approximations are therefore 3.6 and 1.4. For comparison, the exact characteristic polynomial is

$$(2-\lambda)(3-\lambda) - 1 = \lambda^2 - 5\lambda + 5,$$

so the eigenvalues are $(5 \pm \sqrt{5})/2$, approximately 3.61803 and 1.38197. The nonzero off-diagonal entries of A_3 show that three steps have not yet produced an exact diagonalization.

Chapter 12

Singular Value Decomposition

12.1 Singular values and singular vectors

Suppose $A \in \mathbb{R}^{m\times n}$ with $m \geq n$. The matrix A^TA is real symmetric, so it admits an orthonormal eigenbasis $v_1, \ldots, v_n$:

$$A^TAv_i = \lambda_i v_i, \qquad v_i^T v_j = \delta_{ij}.$$

Moreover,

$$0 \leq \|Av_i\|_2^2 = (Av_i)^T(Av_i) = v_i^T A^T A v_i = \lambda_i v_i^T v_i = \lambda_i.$$

Thus all λ_i are nonnegative.

Definition 12.1. The singular values of A are $\sigma_i = \sqrt{\lambda_i}$, arranged so that $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. The vectors v_i are right singular vectors. For $\sigma_i > 0$, define a left singular vector by

$$u_i = \frac{Av_i}{\|Av_i\|_2} = \frac{Av_i}{\sigma_i}.$$

In particular, $Av_i = \sigma_i u_i$, with $v_i \in \mathbb{R}^n$ and $u_i \in \mathbb{R}^m$.

Exercise 12.1. Complete the construction of the singular value decomposition, including the case of zero singular values.

Solution. Let r be the number of positive singular values. For $1 \leq i, j \leq r$,

$$u_i^T u_j = \frac{v_i^T A^T A v_j}{\sigma_i \sigma_j} = \frac{\lambda_j \delta_{ij}}{\sigma_i \sigma_j} = \delta_{ij}.$$

Complete $u_1, \ldots, u_r$ to an orthonormal basis $u_1, \ldots, u_m$ of $\mathbb{R}^m$. If $i > r$, then $\|Av_i\|_2^2 = \lambda_i = 0$, so $Av_i = 0$; no division by σ_i is required. Set

$$U = (u_1 \ \cdots \ u_m), \qquad V = (v_1 \ \cdots \ v_n),$$

and let $\Sigma \in \mathbb{R}^{m\times n}$ have entries $\Sigma_{ii} = \sigma_i$ and all other entries zero. Column by column, $AV = U\Sigma$, hence

$$A = U\Sigma V^T, \qquad U^TU = I_m, \qquad V^TV = I_n.$$

Equivalently,

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$

Indeed, the eigenbasis v_i shows that the nullspace of A is spanned by $v_{r+1}, \ldots, v_n$, so $r = \operatorname{rank}(A)$.

12.2 Worked singular value decompositions

Exercise 12.2. Find a singular value decomposition of

$$A = \begin{pmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{pmatrix}.$$

Solution. Since A is symmetric,

$$A^T A = A^2 = \begin{pmatrix} 5/2 & -3/2 \\ -3/2 & 5/2 \end{pmatrix}.$$

Its characteristic polynomial is

$$(5/2 - \lambda)^2 - 9/4 = (\lambda - 1)(\lambda - 4).$$

For $\lambda = 1$ and $\lambda = 4$, the respective nullspace equations are

$$\begin{pmatrix} 3/2 & -3/2 \\ -3/2 & 3/2 \end{pmatrix} v = 0, \qquad \begin{pmatrix} -3/2 & -3/2 \\ -3/2 & -3/2 \end{pmatrix} v = 0.$$

Choose normalized eigenvectors

$$v_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad v_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Direct multiplication gives

$$Av_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = v_+, \qquad Av_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2v_-.$$

Thus the corresponding left singular vectors equal the right singular vectors, and a factorization in this order is

$$A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

To arrange singular values in decreasing order, interchange both pairs of singular vectors:

$$U = V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad A = U\Sigma V^T.$$

Exercise 12.3. For real parameters a, b, find the singular values and a singular value decomposition of

$$A = \begin{pmatrix} a & a \\ -a & -a \\ b & -b \\ -b & b \end{pmatrix}.$$

Solution. The two columns have squared norm $2a^2 + 2b^2$ and inner product $2a^2 - 2b^2$. Consequently

$$A^T A = \begin{pmatrix} 2a^2 + 2b^2 & 2a^2 - 2b^2 \\ 2a^2 - 2b^2 & 2a^2 + 2b^2 \end{pmatrix},$$

$$\begin{aligned} \det(A^T A - \lambda I) &= (2a^2 + 2b^2 - \lambda)^2 - (2a^2 - 2b^2)^2 \\ &= (4a^2 - \lambda)(4b^2 - \lambda). \end{aligned}$$

Thus the singular values, before ordering, are $2|a|$ and $2|b|$. The corresponding orthonormal right singular vectors are

$$v_a = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad v_b = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Their images are

$$Av_a = \begin{pmatrix} \sqrt{2}a \\ -\sqrt{2}a \\ 0 \\ 0 \end{pmatrix}, \qquad Av_b = \begin{pmatrix} 0 \\ 0 \\ \sqrt{2}b \\ -\sqrt{2}b \end{pmatrix}.$$

Define $s_a = 1$ if $a \geq 0$ and $s_a = -1$ if $a < 0$, and define s_b similarly. Then $a = s_a|a|$, $b = s_b|b|$. Put

$$U_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} s_a & 0 \\ -s_a & 0 \\ 0 & s_b \\ 0 & -s_b \end{pmatrix}, \qquad \Sigma_0 = \begin{pmatrix} 2|a| & 0 \\ 0 & 2|b| \end{pmatrix}, \qquad V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The columns of U_0 are orthonormal even when a parameter is zero, and

$$U_0 \Sigma_0 V^T = \begin{pmatrix} a & a \\ -a & -a \\ b & -b \\ -b & b \end{pmatrix} = A.$$

In particular, for $a, b > 0$ the nonnegative diagonal entries are $2a, 2b$. A full orthogonal matrix is

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} s_a & 0 & 1 & 0 \\ -s_a & 0 & 1 & 0 \\ 0 & s_b & 0 & 1 \\ 0 & -s_b & 0 & 1 \end{pmatrix}.$$

With

$$\Sigma = \begin{pmatrix} 2|a| & 0 \\ 0 & 2|b| \\ 0 & 0 \\ 0 & 0 \end{pmatrix},$$

we have the full decomposition $A = U\Sigma V^T$. If $|b| > |a|$, exchange the first two columns of U and V and exchange the first two diagonal entries of Σ to obtain decreasing order. If $|a| = |b|$, either order is valid. The rank is the number of nonzero parameters among a, b.